Splitting, Dichotomy, and Composition in Finite Extensional Magmas: Independence, Connecting Axioms, and a Canonical Eight-Element Model

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ABSTRACT

Internal splitting (S), the classifier dichotomy (D), and internal composition (C) are three natural algebraic properties of finite extensional 2-pointed magmas. We prove they are pairwise independent: no one implies any other. Lean-verified finite counterexamples at sizes 4 and 5 establish all six non-implications, each with a provably tight bound. A joint irredundance result strengthens this: all eight Boolean combinations of (S, D, C) are realized, so no property is a Boolean function of the other two. Neither result is a small-size artifact: explicit parametric families realize every non-implication and every Boolean profile at every admissible carrier size. The minimum coexistence witness has $N = 5$ and is optimal, since the ICP requires 3 pairwise distinct core elements.

The setting is fixed by four universal obstructions: any nontrivial total magma with a **k**-combinator must be infinite; associativity is incompatible with combining a classifier and a retraction pair; commutativity is incompatible with two distinct left-absorbers; and a single left-absorber collapses any classifier. What survives is finite, non-associative, non-commutative, extensional 2-pointed magmas.

Further structural results: the three-category decomposition induced by D is an isomorphism invariant; the automorphism group of any $N = 5$ S+D+C magma has order at most 2, with the unique possible non-trivial element being the classifier transposition $(\tau_1\ \tau_2)$; a size-driven phase transition takes the role-shape from a single forced configuration at $N = 5$ (every retraction pair satisfies $s = r$, every C-triple has the form (τ_1, g, τ_2)) to a heterogeneous landscape at $N \geq 6$ where strong S re-opens and multiple shape classes coexist; and the ICP is logically equivalent to the standard Compose+Inert axioms. A connecting-axiom programme follows: sorting (compositionality of the classifier/non-classifier partition) is independent of S+D+C and, with S, admits exactly two worlds — class-preserving and class-swapping; sort-introspection fixes the quotation law in each world; and a canonical eight-element model, the lexicographically minimal member of a law-determined 168-model space, is derived rather than designed — observable quotation forces the swap world and the carrier size (sharp), the eval side of its law set is provably redundant, and reversibility, parity, and orbit closure are free, leaving a finite, named residue of definitions and tie-breaks. Lean-verified results — independence counterexamples with tight bounds, joint-irredundance witnesses, decomposition invariance, the ICP equivalence, the all-N scaling witnesses (coexistence, all six non-implications, and all eight cube cells), C-triple type-coherence, the $N = 5$ structure theorem, the mirror-row theorem, and the frame-derivation and law-reduction theorems — compile with zero `sorry`; the remaining universal-N algebraic results in Section 3 not yet transcribed to Lean have hand proofs supported by Z3 UNSAT certificates.

KEYWORDS

finite extensional magma, pairwise independence, classifier dichotomy, internal composition, sorted magmas, canonical model, Lean 4, machine-checked proofs

1 INTRODUCTION

A finite magma is the Cayley table of a binary operation, stripped of syntax and types. We study a particular kind: *finite extensional algebras with binary judgment* — finite magmas whose binary operation returns, alongside data outputs, judgments drawn from a fixed two-element subset of the carrier playing the role of truth values. Within this setting we ask three natural operational questions: can the algebra *represent* its own data, can it *classify* its own structure, and can it *compose* its own operations. Each question has a precise formal answer (S, D, C), and our main result is that the three are pairwise independent: no one implies any other.

The setting is fixed by three structural commitments, all of them positive. *Extensionality* is the foundational one: an element's identity is exhausted by its row in the Cayley table, so an element *is* its action under the operation [6]. *Two distinguished truth values* z_1, z_2 are the minimum number for the operation to register a binary outcome; equipping the algebra with judgment cardinality one is degenerate (Remark 1), and higher arities are natural generalisations we do not pursue. *Inertness of the truth values* formalises what makes them truth values rather than data: $z_i \cdot x = z_i$ for all x, so post-judgment behaviour depends only on judgment status. The truth values are operationally distinguishable (they absorb), structurally interchangeable (no axiom prefers z_1 to z_2), and exhaustive among inert elements (no third element is left-absorbing). Together these commitments select the class of finite extensional 2-pointed magmas.

The three capabilities. Let $\mathrm{core}(S) = S \setminus \{z_1, z_2\}$ denote the carrier minus its truth values. *Internal splitting (S)*: there exists a section/retraction pair $(s, r) \in \mathrm{core}(S)^2$ with $r \cdot (s \cdot x) = x$ on $\mathrm{core}(S)$, so the algebra has internal encoding/decoding for its own data. *Internal dichotomy (D)*: every element of core has the same judgment behaviour across all of core — either every output lies in $\{z_1, z_2\}$ (a *classifier*) or no output does (a *non-classifier*) — and at least one example of each type exists. *Internal composition (C)*: there exist

Preprint version. The Lean proofs, SAT search scripts, and manuscript text were developed with assistance from Claude (Anthropic). All results have been independently verified: Lean proofs compile with `lake build` (zero `sorry`), and SAT counterexamples are frozen in the repository for reproducibility. The author takes full responsibility for all contents.

three pairwise distinct core elements a, b, c with b core-preserving and $a \cdot x = c \cdot (b \cdot x)$ on $\mathrm{core}(S)$, with a taking at least two distinct values — a non-trivial factorization in the left-regular representation. Each capability is what an external machine would otherwise have to supply (naming and decoding, binary judgment, operation chaining), realised instead as a behaviour of elements of the algebra modulo the operation.

Three structural results. Pairwise independence (Section 6): no one of the three capabilities implies any other. Lean-verified finite counterexamples at sizes $N \in \{4, 5\}$ establish all six non-implications, each at its provably minimal size, and explicit parametric families realize every non-implication at every size $n \geq 5$ (Theorem 6.2). Joint irredundance (Section 6.1): all eight Boolean combinations of (S, D, C) are realised by Lean-verified witnesses — again at every admissible size — so no capability is a Boolean function of the other two. Optimal coexistence (Section 7): the smallest cardinality at which all three capabilities can hold simultaneously is $N = 5$, and this is tight: ICP requires three pairwise-distinct core elements, forcing $|S| \geq 5$. Each Cayley table is found by SAT search and independently verified in Lean 4 [9] with zero `sorry`; the parametric families are verified by symbolic case analysis, uniformly in n.

What follows from binary judgment. Several classical algebraic identities are incompatible with this setting. The proofs share a single schema: any equation whose specialisation at a classifier τ forces τ's row to be constant collides with extensionality, equating τ with one of the truth values. From D alone: commutativity, mediality, right self-distributivity all fail (Theorems 3.4, 3.5, 3.6). From S+D: associativity, left self-distributivity, right identity all fail (Theorems 3.10, 3.11, 3.9). The classical algebraic hierarchy — semigroups, monoids, groups, rings — therefore cannot host all three capabilities, despite each capability being individually expressible in classical terms. The intuition behind the schema is that binary judgment is intrinsically non-chainable: composing the operation through a classifier collapses the judgment to a fixed verdict, eliminating discriminative content.

What lies beyond. Two structural results extend the picture. The three-category decomposition $S = Z \sqcup C \sqcup N$ induced by D is an isomorphism invariant (Theorem 4.2); the unique coarsest nontrivial isomorphism-invariant partition of core is the classifier/nonclassifier split (Corollary 4.3). At $N = 5$ a structure theorem (Theorem 4.8) pins the role assignments of retraction, classifier, and C-triple onto the three core slots with no slack: every retraction pair has $s = r$, every C-triple has the form (τ_1, g, τ_2), and the only possible non-trivial automorphism is the classifier transposition $(\tau_1\ \tau_2)$ (Theorem 4.11). At $N \geq 6$ strong S re-opens and the role-shape fractures (Proposition 4.12), but partial invariants survive across the transition (Propositions 4.13, 4.14).

Relation to universality programmes. Combinatory algebras (PCAs, TCAs) [10–12] and Turing categories [14] solve a different problem: full internal universality of the operation, paid for by infinite cardinality (Theorem 3.14, the K-infinity result of the PCA tradition). The setting of this paper retains finite cardinality and obtains, in exchange, a weaker form of universality — representation without a global projector, classification without a subobject classifier, composition without a total internal hom. The three capabilities S, D, C are what survives the trade. Section 9 discusses the finite-vs-infinite cardinality dichotomy in more detail; it is context for our setting, not a fact derivable inside it.

Contributions.

(1) A positive axiomatisation of the setting as finite extensional algebras with binary judgment, together with a uniform classifier-constancy schema explaining why associativity, mediality, self-distributivity, right identity, and commutativity all fail (Section 3).
(2) An independence theorem: S, D, and C are pairwise independent, with Lean-verified counterexamples in all six directions, each at its provably minimal size ($N = 4$ or $N = 5$), together with parametric families realizing every non-implication at every size $n \geq 5$ (Section 6), and coexistence witnesses at $N = 5$ and $N = 6$ (Section 7).
(3) An optimal bound: $N = 5$ is the minimum carrier size for S+D+C coexistence; ICP is impossible at $N = 4$ by pigeonhole (Theorem 7.1).
(4) Decomposition invariance and role rigidity: the $Z/C/N$ partition is preserved by isomorphisms (Section 4); the $N \geq 6$ principal coexistence witness and the smaller S+D witnesses have trivial automorphism group, while the canonical $N = 5$ S+D+C witness internalises a classifier-swap as the action of its non-classifier (Proposition 4.6, Remark 10); at $N = 5$ a structure theorem forces retraction, classifier, and C-triple onto the three core slots and rules out strong S, and the mirror-row theorem (Theorem 4.11) bounds $|\mathrm{Aut}(M)| \leq 2$; at $N \geq 6$ strong S returns and rigidity fails (Proposition 4.12).
(5) A characterization of C: the ICP is logically equivalent to the standard Compose+Inert axioms under variable renaming, with Lean-verified necessity of the non-degeneracy guard conditions (Section 5).
(6) Joint irredundance: all eight cells of the (S, D, C) Boolean cube are inhabited by Lean-verified witnesses, at every admissible carrier size (Section 6.1, Theorem 6.2), so no capability is a Boolean function of the other two — at any size.

2 DEFINITIONS

Throughout, $(S, \cdot)$ is a finite magma on $\mathrm{Fin}(n) = \{0, 1, \ldots, n-1\}$ with binary operation $\cdot : \mathrm{Fin}(n) \times \mathrm{Fin}(n) \to \mathrm{Fin}(n)$.

Definition 2.1 (Finite Extensional Algebra with Binary Judgment). A *finite extensional algebra with binary judgment* — equivalently, an *extensional 2-pointed magma* — is a tuple $(S, \cdot, z_1, z_2)$ where:

(1) $z_1, z_2 \in S$ are distinct elements designated as the *truth values* of the algebra.
(2) The truth values are *operationally inert*: $z_i \cdot x = z_i$ for all $x \in S$. Equivalently, z_1 and z_2 are left-absorbers.
(3) No other element is operationally inert: no other element is a left-absorber.
(4) *Extensionality*: distinct elements have distinct rows in the Cayley table.

We write $\mathrm{core}(S) := S \setminus \{z_1, z_2\}$ for the elements of S that are not truth values.

The four clauses isolate four structural commitments. Clause (i) declares two distinguished judgment outputs and requires them to be distinct — "binary" means two-valued. Clause (ii) gives the truth values their operational meaning: their post-judgment behaviour depends only on their judgment status, not on what they are judging. Clause (iii) closes the truth-value subset against operational impostors: every other element of S varies under left-multiplication, so no other element "acts like a truth value" without being one. Clause (iv) is the foundational extensional commitment: an element's row in the Cayley table is a complete operational description of it, so equality of behaviour is equality.

The choice of cardinality two for the judgment subset is forced by non-degeneracy:

Remark 1 (Why binary, not unary). If a candidate algebra had only one truth value z, any element τ purportedly classifying inputs against $\{z\}$ would satisfy $\tau \cdot x = z$ for all x. The row of τ would then equal the row of z (both constantly z), and extensionality would force $\tau = z$ — the supposed classifier collapses into the truth value itself. Binary judgment is therefore the smallest non-degenerate case, not a parameter choice; k-valued generalisations for $k \geq 3$ strengthen the theory in a different direction we do not pursue here. Lean: `single_absorber_collapse`.

The clause (iii) requirement that no other element be a left-absorber is independent of clauses (i), (ii), and (iv) together with the classifier dichotomy of Definition 2.4: at every $N \in \{4, 5, 6\}$, Z3 produces an extensional 2-pointed magma satisfying D in which a third core element has a constant row (script `scripts/canonicality/no_other_zeros_independence.py`). The clause is therefore a genuine axiom, not a consequence of the rest.

We do *not* commit to a stronger reading of the two truth values as "opposites" in any structural sense. The theorems are compatible both with a symmetric reading (two indistinguishable judgment outputs) and a polar reading (two poles of a polarity). Whether z_1 corresponds to "false" and z_2 to "true" or vice versa is a free choice the axioms do not constrain. This freedom is genuine: applications can specialise the reading without changing the theorems.

Definition 2.2 (Faithful Retract Magma). A *Faithful Retract Magma* (FRM) extends an extensional 2-pointed magma with a *retraction pair* $s, r \in \mathrm{core}(S)$ satisfying $r \cdot (s \cdot x) = x$ for all $x \in \mathrm{core}(S)$, with $r \cdot z_1 = z_1$ (anchoring). The remaining absorber outputs of s and r are unconstrained.

Retraction pair convention. Throughout this paper the retraction pair (s, r) satisfies the mutual-inverse condition on core: $r \cdot (s \cdot x) = x$ and $s \cdot (r \cdot x) = x$ for all $x \in \mathrm{core}(S)$. This matches the Lean formalisation. The choice is load-bearing, not stylistic. One-sided retraction ($r \cdot (s \cdot x) = x$) is a decodable-encoding condition: s produces codes that r can recover. Mutual inverse is the stronger, faithful-representation condition: s and r invert each other in both directions on core, so the encoding neither invents nor loses information. Most theorems (9 of 11) go through under the one-sided hypothesis, and we note this where relevant; two results, retraction pair placement (Theorem 3.7) and the tight bound $|S| \geq 5$ when $s \neq r$ (Theorem 3.8), genuinely require the full mutual inverse. Proposition 2.3 shows this distinction is non-vacuous.

Proposition 2.3 (One-Sided / Mutual-Inverse Separation). *There is an extensional 2-pointed magma on* $\mathrm{Fin}(4)$ *admitting a one-sided retraction pair* (s, r) *with* $s \neq r$, $s, r \in \mathrm{core}$, $r \cdot z_1 = z_1$, *and* $r \cdot (s \cdot x) = x$ *for all* $x \in \mathrm{core}$, *but no mutual-inverse pair* (s', r') *with* $s' \neq r'$. *The explicit Cayley table (rows indexed by the left operand) is:*

$\cdot$	0	1	2	3
0	0	0	0	0
1	1	1	1	1
2	1	0	1	3
3	0	2	1	3

with $z_1 = 0$, $z_2 = 1$, $s = 2$, $r = 3$. *Mutual inverse fails at* $x = 2$: $s \cdot (r \cdot 2) = 2 \cdot 1 = 0 \neq 2$. *Lean:* `mutual_strictly_stronger_than_one_sided`.

Proof. All axioms are verified by direct inspection; Lean discharges each by `decide`, including the non-existence of a mutual-inverse pair with $s' \neq r'$ over $\mathrm{Fin}(4) \times \mathrm{Fin}(4)$. The separation establishes that mutual inverse strictly implies one-sided retraction, so the choice of axiom is substantive, and that the tight bound of Theorem 3.8 cannot be weakened to one-sided. □

The retraction pair gives the algebra a section/retraction structure on core: s acts as a section and r as a retraction. We do not assume a canonical global encoding.

Definition 2.4 (Classifier Dichotomy). An extensional 2-pointed magma satisfies the *classifier dichotomy* when:

(1) A *classifier* $\tau \in \mathrm{core}(S)$ exists with $\tau \cdot x \in \{z_1, z_2\}$ for all $x \in S$.
(2) For every $y \in \mathrm{core}(S)$, either
   - $y \cdot x \in \{z_1, z_2\}$ for all $x \in \mathrm{core}(S)$ — y is a *classifier*, or
   - $y \cdot x \notin \{z_1, z_2\}$ for all $x \in \mathrm{core}(S)$ — y is a *non-classifier*.
(3) At least one non-classifier exists (the dichotomy is not one-sided).

The three clauses split into structural and non-vacuity content. Clauses (i) and (ii) together, D_{struct}, are the structural fragment: the algebra contains *some* classifier and the classification is global. Clause (iii) ensures both classes are inhabited. In the joint S+D+C setting, clause (iii) is automatic: the middle element b of any C-triple is core-preserving (Definition 2.5 clause 1), hence a non-classifier, hence supplies the missing witness (Lean: `DStructRetractMagma.has_non_clas`
Theorem 3.3 below makes the split formal: $D \iff D_{\mathrm{struct}} \wedge P$, where P asserts the existence of a core-preserving element.

The dichotomy creates a three-category decomposition $S = Z \sqcup C \sqcup N$ where $Z = \{z_1, z_2\}$ (truth values), C (classifiers), and N (non-classifiers). No element can be partially classifying. We call outputs in $\{z_1, z_2\}$ *absorber-valued.*

Definition 2.5 (Internal Composition Property). A magma $(S, \cdot, z_1, z_2)$ satisfies the *Internal Composition Property* (ICP) when:

$$\exists\, a, b, c \in \mathrm{core}(S),\ \text{pairwise distinct, such that:}$$

(1) *Core-preservation*: $b \cdot x \notin \{z_1, z_2\}$ for all $x \in \mathrm{core}(S)$.
(2) *Factorization*: $a \cdot x = c \cdot (b \cdot x)$ for all $x \in \mathrm{core}(S)$.
(3) *Non-triviality*: $|\{a \cdot x : x \in \mathrm{core}(S)\}| \geq 2$.

ICP says that the left-regular representation contains a non-trivial factorization on core: one core action factors through two others, one of them core-preserving — a single fragment of internal composition. The pairwise-distinctness and non-triviality clauses are guard conditions blocking degeneracies; both are proved necessary by Lean-verified witnesses (Section 5). In Section 5 we prove that ICP is equivalent to the standard operational axioms Compose+Inert (Theorem 5.1).

Definition 2.6 (The Three Capabilities). An extensional 2-pointed magma has:

- Capability S (*internal splitting*) if it admits a retraction pair (Definition 2.2).
- Capability D (*internal dichotomy*) if it satisfies the classifier dichotomy (Definition 2.4).
- Capability C (*internal composition*) if it satisfies the Internal Composition Property (Definition 2.5).

All three capabilities are properties of the same base structure. None is defined as an extension of any other, making the independence theorem a statement about six non-implications, all proved by Lean-verified counterexamples.

Why these three. Each capability answers a specific positive question about a finite extensional algebra with binary judgment, in a way that is operational rather than typed. Extensionality identifies each element with its row in the Cayley table, so the elements of the algebra are a family of typed maps $\{L_a^{\text{core}} : \text{core} \to \text{core} \cup Z\}$, one per element, with the truth-value maps L_{z_1}, L_{z_2} being constant. S, D, and C correspond to three fundamental structural questions about such a family:

- *Splitting* (S): does the identity on core split through the family — do some L_s, L_r satisfy $L_r \circ L_s = \text{id}_{\text{core}}$? This is the standard section-retraction pair [3].
- *Codomain purity* (D): does each map hit core or Z, never both? This is the unique coarsest non-trivial isomorphism-invariant partition of core (Corollary 4.3).
- *Factorization* (C): does the family contain a non-trivial composition $L_a = L_c \circ L_b$ with L_b core-preserving and L_a non-constant? This is the weakest factorization condition that is not vacuously satisfiable (Lean-verified separations in `ICP.lean`).

Each property is minimal in its own sense. S is the standard notion of splitting. D is the coarsest invariant partition. C is the weakest non-degenerate factorization. The independence and joint-irredundance results (Theorems 6.1, 6.3) confirm these are three questions, jointly irredundant.

Other structural questions. The slice (S, D, C) is one choice within a broader axiom landscape: a Z3 canonicality probe tested thirteen candidate operational axioms against S, D, C and found ten independent of the triple in both directions, three collapsing into derived consequences. What distinguishes the (S, D, C) slice — and the precise sense in which it is structurally selectable — is taken up in Section 10.

The internalisation frame. The magma operation $\cdot$ is taken as given throughout — the structuring relation of $(A, \cdot)$, not itself an object inside the structure. This is not a gap but a structural commitment: no finite algebra can contain its own operation as an element without generating a meta-operation, a Tarski/Lawvere-style regress. We study what *can* be internalised within $(A, \cdot)$ modulo $\cdot$: services that would otherwise be supplied externally (naming and decoding, binary judgment, operation chaining), realised instead as behaviours of elements relative to the fixed operation. The three capabilities S, D, C are each one such internalisation; the operation $\cdot$ is the one component deliberately left outside the internalisation catalog.

3 UNIVERSAL THEOREMS

The following theorems are proved by pure algebraic reasoning (no `decide`) and formalized in `Dichotomic.lean` and `NoCommutativity.lean`. We state the hypotheses explicitly: some require only D, while others use both S and D.

3.1 Theorems from D alone

THEOREM 3.1 (THREE-CATEGORY DECOMPOSITION). *In any extensional 2-pointed magma satisfying D, every element is a zero, a classifier, or a non-classifier. The classes are pairwise disjoint: in particular, $C \cap N = \emptyset$ (no element is partially classifying).*

THEOREM 3.2 (P-WITNESS IDENTITY UNDER D). *In any extensional 2-pointed magma satisfying D, an element $y \in \text{core}(S)$ is a P-witness if and only if y is a non-classifier.*

PROOF. ($\Rightarrow$) Let y be a P-witness, so $y \cdot x \in \text{core}(S)$ for all $x \in \text{core}(S)$. Since $\text{core}(S) \cap \{z_1, z_2\} = \emptyset$, we have $y \cdot x \notin \{z_1, z_2\}$ for all $x \in \text{core}(S)$. Hence y is a non-classifier.

($\Leftarrow$) Let y be a non-classifier, so $y \cdot x \notin \{z_1, z_2\}$ for all $x \in \text{core}(S)$ by Definition 2.4. Since $y \cdot x \in S = \{z_1, z_2\} \cup \text{core}(S)$ and $y \cdot x \notin \{z_1, z_2\}$, we have $y \cdot x \in \text{core}(S)$. Hence y is a P-witness. □

Remark 2. Theorem 3.2 equates the P-witness set with the non-classifier set, so the count of P-witnesses equals the count of non-classifiers in any magma satisfying D. At $N = 5$ under S+D+C, the structure theorem (Theorem 4.8) forces exactly one non-classifier, so the P-witness set has size 1, and that single element absorbs all P-related roles — it is simultaneously s, r, and the middle element b of the C-triple. At $N \geq 6$ the non-classifier count can be ≥ 2 (Proposition 4.12), so P-witnesses distribute across multiple elements and role separation becomes available. This identifies one structural mechanism behind the $N = 5 \to N = 6$ phase transition.

THEOREM 3.3 (D-FACTORIZATION THROUGH P). *Let D_{struct} denote the conjunction of dichotomy on $\text{core}(S)$ and existence of a full classifier τ (i.e., D minus the non-classifier-existence clause). For any extensional 2-pointed magma at any finite N, $D \iff D_{\text{struct}} \wedge P$.*

PROOF. Both directions are direct corollaries of Theorem 3.2.

($\Rightarrow$) D gives D_{struct} together with a non-classifier y. By Theorem 3.2, y is a P-witness, so P holds.

($\Leftarrow$) D_{struct} supplies dichotomy on $\text{core}(S)$ and a full classifier τ; P supplies a witness $y \in \text{core}(S)$ with $y \cdot x \in \text{core}(S)$ for all $x \in \text{core}(S)$. By Theorem 3.2, y is a non-classifier. Hence all three D clauses hold. □

THEOREM 3.4 (ASYMMETRY). *No extensional magma with two distinct left-absorbers is commutative.*

THEOREM 3.5 (NO MEDIALITY). *No extensional 2-pointed magma carrying a classifier is medial: the identity $(a \cdot b) \cdot (c \cdot d) = (a \cdot c) \cdot (b \cdot d)$ cannot hold universally. Setting $a = c = \tau$ and $b = d$ collapses both* *sides via the classifier-plus-absorber identity, yielding $\tau \cdot b = \tau \cdot \tau$ for all b; extensionality then forces $\tau \in \{z_1, z_2\}$. Lean:* `no_mediality`.

Theorem 3.6 (No Right Self-Distributivity). *No extensional 2-pointed magma carrying a classifier satisfies right self-distributivity: the identity $(a \cdot b) \cdot c = (a \cdot c) \cdot (b \cdot c)$ cannot hold universally. The proof instantiates the same pattern as Theorem 3.5: with $a = \tau$, both sides reduce to absorber values under τ's classifier property, forcing τ's row constant. Lean:* `no_right_self_distributivity`.

3.2 Theorems from S+D

Theorem 3.7 (Retraction Pair Placement). *In any extensional 2-pointed magma satisfying both S and D, any witness r to S is a non-classifier: $r \in N$. (Under the stronger mutual-inverse condition in the Lean formalization, $s \in N$ also holds.)*

Theorem 3.8 (Cardinality Bounds). $|S| \geq 4$ *for any extensional 2-pointed magma satisfying both S and D. (The bound $|S| \geq 4$ is nearly immediate from the definitions, since the four roles (two absorbers, one classifier, one non-classifier) must be distinct; the Lean-verified proof confirms that no subtle interaction reduces it. The bound is tight: a 4-element witness with $s = r$ is exhibited in Appendix A.) Under the stronger mutual-inverse condition with $s \neq r$, $|S| \geq 5$ (also tight); this is the nontrivial bound, as it shows the retraction pair consumes two distinct core slots.*

Theorem 3.9 (No Right Identity). *No extensional 2-pointed magma satisfying both S and D has a right identity element.*

Theorem 3.10 (No Associativity). *No extensional 2-pointed magma carrying both a classifier and a retraction pair is associative. In particular, no extensional 2-pointed magma satisfying both S and D is a semigroup, monoid, group, or ring: the classifier dichotomy with a retraction pair is incompatible with the entire classical algebraic hierarchy.*

Proof. Assume $(\tau \cdot a) \cdot b = \tau \cdot (a \cdot b)$ for all a, b. We derive a contradiction in four steps. The proof uses only extensionality, the existence of a classifier τ, and a retraction pair (s, r). Neither the global dichotomy nor the non-classifier existence axiom is needed. Lean file: `Dichotomic.lean`.

Step 1: τ absorbs on the right. For any a, b:

$$\tau \cdot (a \cdot b) = (\tau \cdot a) \cdot b = z_i \cdot b = z_i = \tau \cdot a$$

where the second equality uses $\tau \cdot a \in \{z_1, z_2\}$ (classifier), and the third uses left-absorption. So $\tau \cdot (a \cdot b) = \tau \cdot a$ for all a, b.

Step 2: τ is constant on core. For any core x: the retraction pair gives $x = r \cdot (s \cdot x)$. Applying Step 1 with $a = r$, $b = s \cdot x$:

$$\tau \cdot x = \tau \cdot (r \cdot (s \cdot x)) = \tau \cdot r.$$

Set $v = \tau \cdot r \in \{z_1, z_2\}$. Then $\tau \cdot x = v$ for all core x.

Step 3: τ is constant on absorbers. Since τ is a classifier, it hits both z_1 and z_2; otherwise, if $\tau \cdot x = z_1$ for all x, then τ and z_1 have identical rows, so $\tau = z_1$ by extensionality, contradicting $\tau \neq z_1$ (and symmetrically for z_2). Let b_1, b_2 witness $\tau \cdot b_1 = z_1$ and $\tau \cdot b_2 = z_2$. Applying Step 1 with $a = \tau$:

$$\tau \cdot z_1 = \tau \cdot (\tau \cdot b_1) = \tau \cdot \tau = v, \qquad \tau \cdot z_2 = \tau \cdot (\tau \cdot b_2) = \tau \cdot \tau = v.$$

The middle equalities use Step 1; $\tau \cdot \tau = v$ because τ is core (Step 2).

Step 4: Contradiction. Steps 2 and 3 give $\tau \cdot x = v$ for all $x \in S$. Since $v \in \{z_1, z_2\}$, the classifier τ has the same row as absorber v. By extensionality, $\tau = v \in \{z_1, z_2\}$, contradicting $\tau \neq z_1, z_2$. □

Remark 3. The retraction pair is essential in Step 2: the identity $\tau \cdot x = \tau \cdot (r \cdot (s \cdot x)) = \tau \cdot r$ uses $r \cdot (s \cdot x) = x$ to show τ is constant on core. Without a retraction pair, the proof breaks, and the theorem does not extend to "no extensional 2-pointed magma with a classifier is associative."

Theorem 3.11 (No Left Self-Distributivity). *No extensional 2-pointed magma satisfying both S and D satisfies left self-distributivity: the identity $a \cdot (b \cdot c) = (a \cdot b) \cdot (a \cdot c)$ cannot hold universally. The specialization $a = \tau$ yields $\tau \cdot (b \cdot c) = \tau \cdot b$ (Step 1 of the no-associativity proof); the retraction-pair argument (Steps 2–4) then proceeds identically. Lean:* `no_left_self_distributivity`.

The classifier-constancy schema. Theorems 3.4, 3.5, 3.6, 3.10, and 3.11 share a proof schema, summarised in Section 1: specialisation at $a = \tau$ (or $a = c = \tau$) forces τ's row constant, colliding with extensionality. Commutativity, mediality, and right self-distributivity instantiate the schema from D alone; associativity and left self-distributivity use S+D, with the retraction pair extending classifier-constancy from core to absorbers.

3.3 Theorems from D+C

Theorem 3.12 (C-triple type-coherence). *In any extensional 2-pointed magma satisfying both D and C, the outer elements a and c of any C-triple (a, b, c) are either both classifiers or both non-classifiers.*

Proof. Let (a, b, c) be a C-triple, so $a \cdot x = c \cdot (b \cdot x)$ for all $x \in \mathrm{core}(S)$, with $b \cdot x \in \mathrm{core}(S)$ for all $x \in \mathrm{core}(S)$ (core-preservation of b).

Suppose c is a classifier. Then $c \cdot y \in \{z_1, z_2\}$ for all $y \in \mathrm{core}(S)$, so in particular $c \cdot (b \cdot x) \in \{z_1, z_2\}$ for all $x \in \mathrm{core}(S)$. By the C-triple equation, $a \cdot x \in \{z_1, z_2\}$ for all $x \in \mathrm{core}(S)$, so a is also a classifier.

Suppose c is a non-classifier. Then $c \cdot y \notin \{z_1, z_2\}$ for all $y \in \mathrm{core}(S)$, so $c \cdot (b \cdot x) \notin \{z_1, z_2\}$ for all $x \in \mathrm{core}(S)$. By the C-triple equation, $a \cdot x \notin \{z_1, z_2\}$ for all $x \in \mathrm{core}(S)$, so a is also a non-classifier.

Either way, a and c have the same classifier-type. □

Remark 4. Theorem 3.12 is invoked at Step 3 of the $N = 5$ structure theorem (Theorem 4.8) and isolates the universal-N content that the structure theorem then specializes to size 5.

Theorem 3.13 (Classifier-count bound). *In any extensional 2-pointed magma satisfying both D and C, either $|C| \geq 2$ or $|N| \geq 3$, where C and N denote the classifier and non-classifier subsets of $\mathrm{core}(S)$.*

Proof. Let (a, b, c) be any C-triple (existing by C). By Theorem 3.2, b is a non-classifier (since b is core-preserving by clause 1 of Definition 2.5), so $b \in N$. By Theorem 3.12, a and c are either both in C or both in N.

If $a, c \in C$: since $a \neq c$, $|C| \geq 2$.

If $a, c \in N$: since a, b, c are pairwise distinct, $|N| \geq 3$. □

Remark 5. At $N = 5$ the core has exactly 3 elements, so $|C|+|N| = 3$. D requires $|C| \geq 1$ (a full classifier exists), so $|N| \leq 2$, ruling out the $|N| \geq 3$ branch of Theorem 3.13. Hence $|C| \geq 2$, and combined with $|N| \geq 1$ this forces $|C| = 2$ and $|N| = 1$ exactly — recovering the classifier count of the $N = 5$ structure theorem (Theorem 4.8). At $N \geq 6$ both disjuncts become satisfiable: the $|C| \geq 2$ branch is realized by the explicit witness in Appendix A, and the $|N| \geq 3$ branch is realized by the structurally distinguished single-classifier class in the strong-S enumeration of `scripts/enumerate_rdh_iso_N6_strongR.json`.

3.4 Contrast with combinatory algebras: K-infinity

The following classical result, recalled here in self-contained form, frames the contrast between this paper's finite-cardinality setting and the standard PCA/TCA tradition (cf. Section 1): combinatory algebras pay infinite cardinality for full internal universality, while extensional 2-pointed magmas retain finiteness and obtain the partial forms S, D, C.

Theorem 3.14 (K-infinity). *Let $(A, \cdot)$ be a total magma with $|A| \geq 2$ carrying an element $\mathbf{k}$ that satisfies $\mathbf{k} \cdot a \cdot b = a$ for all $a, b \in A$. Then A is infinite. In particular, every nontrivial combinatory algebra is infinite.*

Proof. The map $\varphi(x) = \mathbf{k} \cdot x$ is injective: $\mathbf{k} \cdot a = \mathbf{k} \cdot b$ implies $\mathbf{k} \cdot a \cdot c = \mathbf{k} \cdot b \cdot c$ for all c, hence $a = b$. We claim $\mathbf{k} \notin \mathrm{Im}(\varphi)$. Suppose $\mathbf{k} \cdot x = \mathbf{k}$ for some x. Then for all y, $\mathbf{k} \cdot x \cdot y = \mathbf{k} \cdot y$, so $x = \mathbf{k} \cdot y$ for all y. By injectivity, all y are equal, contradicting $|A| \geq 2$. Thus $\varphi : A \to A$ is an injection missing $\mathbf{k}$, which is impossible for finite A. Lean: `CompletenessWall.lean`, theorems `k_collapse_of_finite` and `no_k_combinator_on_fin` (formalized for arbitrary finite carriers). □

Remark 6. The hypothesis is the $\mathbf{k}$ equation alone: no $\mathbf{s}$ combinator is needed, so the obstruction bites well below full combinatory-algebra structure. Our extensional 2-pointed magmas avoid it: instead of a projector $\mathbf{k}$ with $\mathbf{k} \cdot a \cdot b = a$, they have left-absorbers z with $z \cdot x = z$ (the dual pattern, constant on columns rather than rows) and a retraction pair (s, r) with $r \cdot (s \cdot x) = x$ on core. Neither structure provides a non-surjective injection on the full ground set.

Remark 7 (K-infinity and finite computational substrates). K-infinity shows that no finite carrier admits a total, global $\mathbf{k}$ combinator — the kind of universality that packs all of computation into a single element acting on the rest. It does *not* rule out the weaker notion of a finite algebra serving as the *instruction set* of a universal interpreter whose state and memory are external. A finite Cayley table can in principle function like a CPU's instruction table: every step of a computation is a table lookup, while the evolving state (working memory, stack, continuation) lives outside the algebra as finite sequences of element-indices that grow with the input. In this bounded-memory sense of universality — analogous to a Turing machine whose transition table is finite but whose tape grows with the input — finite extensional 2-pointed magmas are not ruled out as computational substrates. The obstruction bites only when one asks for universality to be *entirely internal* to a single global element of a finite carrier; distributed, stepwise universality over a finite instruction set remains available.

K-infinity is a wall about *size*: finiteness excludes completeness. A second wall, in the opposite direction, shows that completeness excludes the dichotomy — at every cardinality.

Theorem 3.15 (Completeness wall). *No total applicative structure $(A, \cdot)$ — of any cardinality — carrying two left absorbers z_1, z_2 and elements $\mathbf{s}, \mathbf{k}$ satisfying the combinatory equations satisfies the classifier dichotomy. It suffices that every column of the Cayley table be internalized as a row: if for every a there is an element m_a with $m_a \cdot x = x \cdot a$ for all x (the thrush $\lambda x.\, x\, a$, definable as $\mathbf{s}\,(\mathbf{skk})\,(\mathbf{k}\, a)$), then D fails. Neither extensionality nor the distinctness of the absorbers is used.*

Proof. The dichotomy constrains *rows*: every core row is absorber-valued or core-valued on core, never mixed. But D's own axioms provide a mixed *column*: for any classifier τ and non-degeneracy witness $y_0 \cdot x_0 \notin \{z_1, z_2\}$, the column of x_0 contains the absorber-valued entry $\tau \cdot x_0$ and the core-valued entry $y_0 \cdot x_0$. A mixed column is harmless until the structure can transpose. The element $m := m_{x_0}$ has the column of x_0 as its row: $m \cdot \tau = \tau \cdot x_0$ is absorber-valued while $m \cdot y_0 = y_0 \cdot x_0$ is not; moreover m is not itself an absorber (an absorber's row is absorber-valued everywhere, but $m \cdot y_0$ is not). So m is a mixed core element, and the dichotomy fails at m. Lean: `CompletenessWall.lean`, theorems `flip_blocks_dichotomy` and `sk_blocks_dichotomy`, stated over an arbitrary carrier type. □

Remark 8 (The two walls delimit the setting). Together the walls say: finite worlds cannot be combinatorially complete (K-infinity), and complete worlds cannot be dichotomic (Theorem 3.15). In the untyped λ-calculus extended with two absorbing sink constants — or in the $\bar{\lambda}\mu\tilde{\mu}$-calculus of Curien and Herbelin, where absorbers are directly definable as aborting terms $\mu\delta.c$ with δ unused — capabilities S and C are definable for free (Church pairing; the $\mathbf{B}$ combinator), while D provably fails: mixed elements, i.e. "partial deciders" that return absorber values on some core inputs and core values on others, are always definable. The dichotomy is consistent exactly in worlds that cannot internalize their own columns as rows, so the S/D/C landscape of this paper is not a toy model of the λ-calculus but the regime that combinatorial completeness forbids. This sharpens the cross-formalism question of Section 10: the transfer target for a non-vacuous D cannot be any computationally complete calculus; the natural candidates are sub-complete fragments, such as the polarized or focused subsyntaxes of sequent-style calculi, whose producer/consumer discipline enforces syntactically what D demands behaviorally.

Remark 9 (K-infinity and Scott's continuous resolution). Theorem 3.14 is the finite-carrier analogue of the obstruction Scott resolved by moving to continuous lattices and domain theory [7]. Scott's $D \cong [D \to D]$ requires a space whose elements are simultaneously values and functions on values, which is impossible at full function-space cardinality; the resolution is to restrict to continuous functions on an inverse-limit lattice, at the price of countable infinity. Scott's construction retains global self-application and pays the price of infinite cardinality; the E2PM setting retains finite cardinality and pays the price of global self-application. The three capabilities S, D, and C are what survives: representation without a universal projector, classification without a subobject classifier, composition without a total internal hom — the weaker partial forms of the same computational services Scott's construction provides globally.

4 DECOMPOSITION INVARIANCE

Definition 4.1 (Absorber-Preserving Isomorphism). An *absorber-preserving isomorphism* $\varphi : M_1 \to M_2$ between extensional 2-pointed magmas is a bijection $\varphi : \mathrm{Fin}(n) \to \mathrm{Fin}(n)$ satisfying:

- $\varphi(a \cdot_1 b) = \varphi(a) \cdot_2 \varphi(b)$ for all a, b
- $\varphi(z_1^{(1)}) = z_1^{(2)}$ and $\varphi(z_2^{(1)}) = z_2^{(2)}$

Theorem 4.2 (Functoriality). *Any absorber-preserving isomorphism between extensional 2-pointed magmas satisfying D preserves the three-category decomposition: φ maps zeros to zeros, classifiers to classifiers, and non-classifiers to non-classifiers.*

Proof sketch. *Zeros*: if a is a left-absorber, then for all x in the image of φ, $\varphi(a) \cdot_2 \varphi(x) = \varphi(a \cdot_1 x) = \varphi(a)$. By surjectivity, $\varphi(a)$ is a left-absorber.

Classifiers: if $y \cdot_1 x \in \{z_1, z_2\}$ for all core x, then $\varphi(y) \cdot_2 \varphi(x) = \varphi(y \cdot_1 x) \in \{z_1', z_2'\}$ for all core $\varphi(x)$. By surjectivity and injectivity (to exclude absorber preimages), $\varphi(y)$ is a classifier.

Non-classifiers: symmetric argument using injectivity to transport $\notin \{z_1, z_2\}$.

Algebraic proof (no `decide`); Lean file: `Functoriality.lean`. □

Corollary 4.3 (Canonicity of the decomposition). *The C/N partition of core is iso-invariant: every absorber-preserving isomorphism preserves C and N individually (Theorem 4.2). Moreover, it is the unique coarsest non-trivial iso-invariant partition of core, since classifier-status — whether an element's row on core has image contained in $\{z_1, z_2\}$ — is the only intrinsic two-valued algebraic invariant on core.*

Theorem 4.4 (Capability Invariance). *All three properties S, D, and C are preserved by absorber-preserving isomorphisms of extensional 2-pointed magmas: if $\varphi : M_1 \to M_2$ is such an isomorphism and M_1 satisfies S (resp. D, C), then M_2 satisfies S (resp. D, C). Equivalently, S, D, and C are structural properties of the row-image partial monoid $R(M)$ — section-retraction in $R(M)$, codomain decomposition $R(M)|_{\mathrm{core}} = R_{\mathrm{cls}} \sqcup R_{\mathrm{core}}$, and partial composition closure on R_{core} — and so are automatically iso-invariant. Algebraic proof in* `CapabilityInvariance.lean`*; the row-image translation is verified empirically across all 3901 N=5 S+D+C iso classes (*`scripts/row_image_invariants.py`*).*

4.1 Role Rigidity

Theorem 4.2 fixes *role classes*: Z, C, and N are intrinsic invariants of the Cayley table, and Corollary 4.3 adds that C/N is the unique coarsest non-trivial iso-invariant partition of core. Individual role assignments within each class — which absorber is z_1, which classifier is τ, which core element is the section s versus the retraction r — remain, in general, determined only up to $\mathrm{Aut}(M)$.

Definition 4.5 (Role-Rigid Magma). A magma $(S, \cdot)$ is *role-rigid* if the only operation-preserving bijection $\sigma : S \to S$ is the identity — equivalently, $\mathrm{Aut}(M)$ is trivial. Role-rigidity sharpens the class-level canonicity of the $Z/C/N$ partition to the element level: individual role assignments are $\mathrm{Aut}(M)$-invariants of the Cayley table.

Proposition 4.6 (Rigidity and non-rigidity of the paper's principal witnesses). *The following witnesses constructed in this paper are role-rigid:* `dotW6` *(the $N = 6$ S+D+C witness),* `dotK4` *(the $N = 4$ S+D witness),* `dotK5` *(the $N = 5$ S+D witness with $s \neq r$), and* `dotOSS` *(the $N = 4$ one-sided separator of Proposition 2.3). Each is discharged by* `native_decide` *over* Equiv.Perm(Fin(n)) *in* `Rigidity.lean`.

The $N = 5$ S+D+C principal witness `dotW5` *(Appendix A) is* non-role-rigid by design*: it admits the classifier transposition $(\tau_1\ \tau_2)$ as a non-trivial automorphism, and the mirror-row theorem (Theorem 4.11) ensures this is its* only *non-trivial automorphism, so* $|\mathrm{Aut}(\texttt{dotW5})| = 2$. *Lean:* `witness5_classifier_swap_aut` *in* `Rigidity.lean`. *The reason this witness is non-rigid is structural — explained in the canonical-witness remark below.*

Remark 10 (The canonical $N = 5$ witness internalises its symmetry). The $N = 5$ witness `dotW5` is selected by two structural principles, not engineering convenience.

Indicator-style classifiers. Reading absorbers as Boolean truth values (z_1 = FALSE, z_2 = TRUE — which is what D *is* topos-theoretically, with $\Omega = 1+1$), each classifier τ_i is the characteristic function of the singleton subobject $\{\tau_i\}$ extended to fix the truth labels: $\tau_i(x) = z_2$ iff $x \in \{\tau_i, z_2\}$, else z_1. Each classifier identifies itself as TRUE and preserves both absorbers' truth labels.

The non-classifier internalises the classifier swap. The unique non-classifier g acts on core as the transposition $(\tau_1\ \tau_2)$, with $g \cdot g = g$ and g fixing both absorbers. The classifier-swap symmetry guaranteed by Theorem 4.11 is therefore *realised inside the magma* as g's own action on core — the non-trivial automorphism isn't an external symmetry of the Cayley table, it's the map $x \mapsto g \cdot x$ on core.

These two principles together fully determine the Cayley table; this is the unique $N = 5$ S+D+C magma (up to absorber-preserving isomorphism) satisfying both. The cost of this choice is that the witness is non-rigid; the alternative (taking g as the identity on core) would be rigid but would leave the classifier-swap symmetry as a property of the table only, not internalised in the operation. The internalising choice was preferred because it matches the categorical-canonicality reading the paper has been pushing toward, and because it makes the mirror-row theorem's symmetry visibly realised rather than merely possible.

Corollary 4.7 (Polar absorbers in role-rigid magmas). *In a role-rigid magma, no automorphism exchanges the two absorbers: z_1 and z_2 lie in distinct* $\mathrm{Aut}(M)$*-orbits. Lean:* `role_rigid_polar_absorbers`.

Role-rigidity of the paper's specific witnesses is not a general consequence of S+D+C, even at the minimum coexistence size $N = 5$. The role-*shape* (which absorber is z_i, which core element is the unique non-classifier, the canonical (τ_1, g, τ_2) form of every C-triple) is forced universally onto the $Z/C/N$ partition by Theorem 4.8;

element-level rigidity is the strictly stronger condition that no non-trivial automorphism exists. The failure mode is precisely characterised by Theorem 4.11 below: at $N = 5$ the automorphism group is always a subgroup of $\mathbb{Z}/2$, and a magma fails to be role-rigid exactly when the classifier transposition $(\tau_1\ \tau_2)$ is itself a Cayley-table automorphism.

Theorem 4.8 (Structure Theorem at $N = 5$). *Let M be an extensional 2-pointed magma on* $\mathrm{Fin}(5)$ *satisfying S, D, and C, with absorbers z_1, z_2 and core $C = \mathrm{Fin}(5) \setminus \{z_1, z_2\}$ of size 3. Then:*

1. *The core partitions as $C = \{\tau_1, \tau_2, g\}$ where τ_1, τ_2 are classifiers and g is the unique non-classifier.*
2. *Every retraction pair (s, r) in M satisfies $s = r = g$.*
3. *Every C-triple (a, b, c) in M satisfies $b = g$ and $\{a, c\} = \{\tau_1, \tau_2\}$.*

Proof. Write $C = \{2, 3, 4\}$. By D, every core element is either a classifier (row on C lies in $\{z_1, z_2\}$) or a non-classifier (row on C lies outside $\{z_1, z_2\}$), and at least one is a non-classifier. Write k for the number of classifiers in C; then $k \in \{0, 1, 2\}$.

Step 1 (retraction elements are non-classifiers). Let (s, r) be any retraction pair and suppose s is a classifier. Then $s \cdot x \in \{z_1, z_2\}$ for all $x \in C$. Since $|C| = 3 > 2 = |\{z_1, z_2\}|$, pigeonhole gives distinct $x_1, x_2 \in C$ with $s \cdot x_1 = s \cdot x_2$. The retraction identity $r \cdot (s \cdot x_i) = x_i$ then forces $x_1 = x_2$, a contradiction. So s is a non-classifier. The symmetric argument using $s \cdot (r \cdot x) = x$ shows r is a non-classifier.

Step 2 (C's middle element is a non-classifier). Let (a, b, c) be any C-triple (existing by C). By Theorem 3.2, since b is core-preserving (clause 1 of Definition 2.5), b is a non-classifier.

Step 3 (C's outer elements have the same classifier-type). By Theorem 3.12, a and c are either both classifiers or both non-classifiers.

Step 4 (exactly 2 classifiers). By Theorem 3.13, $k \geq 2$ or the non-classifier count in core is ≥ 3. The second alternative would fill the entire 3-element core with non-classifiers, contradicting D's requirement that a full classifier τ exists. Hence $k \geq 2$. Since $|C| = 3$ and D also requires at least one non-classifier, $k \leq 2$. Therefore $k = 2$.

Step 5 (role lock-in). By Step 4, $C = \{\tau_1, \tau_2, g\}$ with g the unique non-classifier. By Step 1, $s, r \in \{g\}$, so $s = r = g$. By Step 2, $b = g$. By Step 3, a and c are both classifiers; since they are distinct, $\{a, c\} = \{\tau_1, \tau_2\}$. □

Theorem 4.8 is machine-checked in Lean 4 (`Magma/StructureN5.lean`): the uniqueness lemma `unique_non_classifier_at_N5` yields `N5_exists_unique_non_classifier` (clause (i)), `N5_sec_eq_ret` (clause (ii), equivalent to Corollary 4.9), and `N5_icp_triple_structure` (clause (iii)). The proof is algebraic (no `decide`, no `native_decide`) and depends only on `propext`, `Classical.choice`, `Quot.sound`. An independent check uses three Z3 UNSAT queries (negating each of (i), (ii), (iii) in turn; `scripts/structure_theorem_N5.py`, each query resolving in under 0.1 s).

Corollary 4.9 (Strong S is absent at $N = 5$). *No S+D+C magma on* $\mathrm{Fin}(5)$ *admits a retraction pair with $s \neq r$. Equivalently, strong S is unsatisfiable in conjunction with S+D+C at size 5.*

Proof. By Theorem 4.8(ii), every retraction pair has $s = r$. □

Corollary 4.10 ($N = 5$ rigidity under strong S). *Every extensional 2-pointed magma on* $\mathrm{Fin}(5)$ *satisfying S with $s \neq r$, D, and C is role-rigid. The statement is vacuously true by Corollary 4.9.*

Theorem 4.11 (Mirror-row theorem at $N = 5$). *Let M be an extensional 2-pointed magma on* $\mathrm{Fin}(5)$ *satisfying S, D, and C, with classifiers τ_1, τ_2 and unique non-classifier g (Theorem 4.8). Every automorphism $\sigma \in \mathrm{Aut}(M)$ fixes both absorbers and the unique non-classifier g, and acts on the two classifiers as either the identity or the transposition $(\tau_1\ \tau_2)$. Consequently:*

1. *$|\mathrm{Aut}(M)| \in \{1, 2\}$;*
2. *the unique possible non-trivial automorphism is the transposition $(\tau_1\ \tau_2)$, extended by identity on $\{z_1, z_2, g\}$;*
3. *M is role-rigid if and only if this classifier transposition is not a Cayley-table automorphism.*

Proof. By Theorem 4.2, σ preserves the $Z/C/N$ partition, so σ permutes $\{z_1, z_2\}$, permutes $\{\tau_1, \tau_2\}$, and fixes g (the unique non-classifier by Theorem 4.8). Hence σ lies in the Klein four-group $\mathrm{Sym}(\{z_1, z_2\}) \times \mathrm{Sym}(\{\tau_1, \tau_2\})$. We rule out the two cases in which σ swaps $\{z_1, z_2\}$.

Case A: σ swaps absorbers and fixes both classifiers. For any classifier τ, $\tau \cdot \tau \in \{z_1, z_2\}$. Applying σ: $\sigma(\tau \cdot \tau) = \sigma(\tau) \cdot \sigma(\tau) = \tau \cdot \tau$, so $\tau \cdot \tau$ is σ-fixed. But σ swaps $\{z_1, z_2\}$. Contradiction.

Case B: σ swaps absorbers and swaps classifiers. Since σ fixes g and g is core-preserving (clause 3 of Theorem 4.8, via Theorem 3.2), $g \cdot g \in \mathrm{core}(S)$ is σ-fixed. Among $\{\tau_1, \tau_2, g\}$ only g is σ-fixed, so $g \cdot g = g$. The factorisation $\tau_1 \cdot x = \tau_2 \cdot (g \cdot x)$ from Theorem 4.8(iii), evaluated at $x = g$, gives $\tau_1 \cdot g = \tau_2 \cdot (g \cdot g) = \tau_2 \cdot g \in \{z_1, z_2\}$. Apply σ: $\sigma(\tau_1 \cdot g) = \sigma(\tau_1) \cdot \sigma(g) = \tau_2 \cdot g$, so $\tau_1 \cdot g$ is σ-fixed. But σ swaps $\{z_1, z_2\}$. Contradiction.

So σ fixes $\{z_1, z_2\}$ pointwise. Together with $\sigma(g) = g$, the only remaining freedom is on $\{\tau_1, \tau_2\}$, which is either fixed or swapped. □

Theorem 4.11 is machine-checked in Lean 4 (`N5_aut_preserves_absorbers` in `Magma/MirrorRow.lean`, zero `sorry`; depends only on `propext`, `Classical.choice`, `Quot.sound`). The phase-cartography enumeration of 3901 $N = 5$ S+D+C iso classes finds exactly 12 admitting the classifier transposition (`scripts/phase_cartography_N5.json`, `scripts/mirro`; these are precisely the non-role-rigid cases.

Proposition 4.12 (Non-rigidity at $N \geq 6$ under strong S). *At $N = 6, 7, 8$ there exist finite extensional 2-pointed magmas satisfying S (mutual-inverse with $s \neq r$), D, and C with non-trivial automorphism group. Z3-verified witnesses with independent Python confirmation:* `scripts/nonrigid_rdh_N{6,7,8}_strongR.json`. *The $N = 8$ witness admits an absorber-swap automorphism, so Corollary 4.7 does not extend to S+D+C under strong S at this size without further hypotheses.*

Proposition 4.13 ($N = 6$ automorphism constraints under strong S). *Every automorphism of an extensional 2-pointed magma on* $\mathrm{Fin}(6)$ *satisfying strong S, D, and C permutes $\{z_1, z_2\}$, and no such magma admits an automorphism that fixes both absorbers and has order 4 on core. Z3 UNSAT:* `scripts/canonicality/nonrigid_characterizatio`

Proof. Both claims are encoded as Z3 UNSAT queries over $\mathrm{Fin}(6)$. The absorber-permutation constraint reduces to checking that no model exists with a non-trivial automorphism σ satisfying $\sigma(z_1) \notin \{z_1, z_2\}$; the order-4 constraint enumerates the absorber-fixing 4-cycles on core (algebraically there are six, and each is asserted not to be an automorphism of T). Each query is independent of any specific witness table. The first query is a strict proof; the second confirms that the non-rigid landscape under strong S at $N = 6$ avoids one structurally distinguished cycle type. The enumeration of non-rigid iso classes at $N = 6$ strong S produces a large landscape — at least 5,985 iso classes were observed before the iterative-blocking solver time-per-class grew past feasibility — dominated by absorber-fixing involutions with $|\mathrm{Aut}| = 2$. The enumerator file `scripts/enumerate_rdh_nonrigid_iso.py` is the source of record; the exact total is not used elsewhere in the paper. □

Proposition 4.14 ($N = 6$ C-triple bound under strong S). *Every extensional 2-pointed magma on* $\mathrm{Fin}(6)$ *satisfying strong S, D, and C admits at least two distinct C-triples* $(a, b, c) \in \mathrm{core}(S)^3$. *The bound is tight: a strong-S+D+C magma on* $\mathrm{Fin}(6)$ *with exactly two C-triples is Z3-constructible. Z3 verification:* `scripts/canonicality/structure_theorem_N6.py`.

Theorem 4.8 and Propositions 4.12–4.14 together sharpen the phase transition picture. At $N = 5$ the retraction pair, the C-triple, and the classifier pair are forced onto the three core slots with no room to spare, and in particular the retraction collapses to $s = r$; strong S is not merely consistent with rigidity but outright unavailable. At $N = 6$, adding one core element re-opens strong S, and with it the possibility of non-trivial automorphisms: both the classifier count and the retraction structure admit choices, and neither the structural lock-in of (i)–(iii) nor the rigidity of the principal witnesses survives. What does survive across the transition: every automorphism still permutes $\{z_1, z_2\}$, the absorber-fixing 4-cycle is forbidden, and at least two distinct C-triples persist (Propositions 4.13, 4.14). The collapse from N=5's tight role lock-in to N=6's flexibility is partial, not total.

A common substructure: core-preserving rows. Let P (*internal preservation*) denote the property: $\exists y \in \mathrm{core}(S)$ with $y \cdot x \in \mathrm{core}(S)$ for all $x \in \mathrm{core}(S)$. Both D and C imply P. D supplies a non-classifier whose row on core lies entirely in core (the all-out branch of the dichotomy), so the non-classifier itself witnesses P; C supplies the middle element b of the factorization triple, which is core-preserving by clause 1 of Definition 2.5. We name P but do not promote it to a fourth capability: because P is implied by both D and C separately, it is not pairwise-independent of them and so does not stand on the same footing as S, D, and C with each other. It is the cleanly-named common substructure they share. The two implications are qualitatively different. Writing D_{struct} for the dichotomy on core plus existence of a full classifier τ (i.e., D minus the existential "at least one non-classifier" clause), and C_{eq} for the factorization equation plus pairwise-distinctness and non-triviality (i.e., C minus core-preservation of b), Theorem 3.3 gives the D-side universally, and Z3 verification at $N \in \{5, 6\}$ gives the C-side:

$$D \iff D_{\mathrm{struct}} \wedge P \quad (\text{any finite } N), \qquad C \Rightarrow P \wedge C_{\mathrm{eq}} \text{ but not conversely.}$$

The structural reading: P is a common substructure, but the way it sits inside D and C differs. In D it is *decoupled* from τ (the witness can be a different element); in C it is *coupled* to the factorization's middle slot b (the witness must be the same element). Verification: `scripts/canonicality/factorization_D_H.py`.

5 ICP CHARACTERIZATION

An algebra with S can encode, decode, and look up any entry in the Cayley table via the retraction pair (s, r). But this requires an *external* machine to supply control flow: dispatch, composition, recursion. C internalizes a non-trivial composition pattern that would otherwise be supplied externally: dedicated elements whose left-multiplication implements composition ($\eta = \rho \circ g$) and storage (g). The ICP captures exactly this: one core action factors through two others, meaning the algebra contains a fragment of its own control flow as an element.

The standard formulation uses two operational axioms:

- *Compose*: $\eta \cdot x = \rho \cdot (g \cdot x)$ for all $x \in \mathrm{core}(S)$
- *Inert*: $g \cdot x \notin \{z_1, z_2\}$ for all $x \in \mathrm{core}(S)$

with η, g, ρ pairwise distinct non-absorbers and η non-trivial (≥ 2 distinct values on core).

Theorem 5.1 (ICP Equivalence). *For any finite magma on a 2-pointed set* $(S, \cdot, z_1, z_2)$ *of any size n:*

$$\textit{ICP} \iff \textit{Compose} + \textit{Inert (with non-trivial composite).}$$

Proof. The bijection between witness triples is $(\eta, g, \rho) = (a, b, c)$: the composed element a is η, the core-preserving element b is g, and the outer element c is ρ. Under this identification the factorization $a \cdot x = c \cdot (b \cdot x)$ becomes $\eta \cdot x = \rho \cdot (g \cdot x)$ (Compose), core-preservation of b becomes Inert, and the remaining conditions (pairwise distinct, non-absorber) transfer directly.

The pairwise-distinctness and non-triviality conditions are not cosmetic; they block two degeneracies, and both are proved necessary by Lean-verified witnesses (`ICP.lean`). Without pairwise-distinctness, the retraction pair (s, r) with $s = r$ trivially satisfies factorization via $s \cdot x = s \cdot (s \cdot x)$, giving every FRM a spurious ICP; the minimal S+D model at $N = 4$ (Cayley table in Appendix A; Lean name `kripke4`) separates the weakened definition from the full one (theorem `icp_distinct_necessary`). Without non-triviality, any constant-on-core element a (with $a \cdot x = v$ for all core x) trivially factors through any core-preserving b by taking c with $c \cdot y = v$; a custom 6-element FRM separates the weakened definition from the full one (theorem `icp_nontrivial_necessary`, `ICP.lean`).

The proof is pure equational logic, with no `decide`, no case analysis on n. It holds for every finite magma on a 2-pointed set, of any size. Lean file: `ICP.lean`, theorem `icp_iff_composeInert`. □

This equivalence gives C a single-concept definition: the left-regular representation contains a non-trivially composed element. The operational axioms (Compose, Inert) and the ICP are the same property under variable renaming. Additionally, ICP is invariant under FRM isomorphisms (algebraic proof, no `decide`; `ICP.lean`, theorem `FRMIso.preserves_icp`).

6 INDEPENDENCE

Theorem 6.1 (Full Independence). *No capability implies any other. Specifically, all six pairwise non-implications hold:*

1. $S \not\Rightarrow D$*: An extensional 2-pointed magma of size* 5 *satisfying S contains a classifier, a pure non-classifier, and a mixed element violating the dichotomy. This is tight: a non-vacuous dichotomy failure requires three pairwise distinct core elements, so* $N \geq 5$.

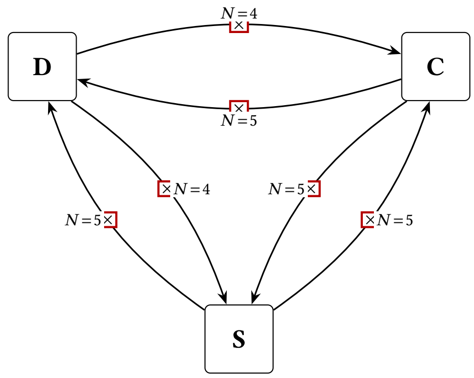

Figure 1: Full independence structure. All six pairwise non-implications are proved by Lean-verified finite counterexamples at the indicated sizes. No capability implies any other.

(2) *$S \not\Rightarrow C$: An extensional 2-pointed magma of size 5 satisfying S has exactly the 3 core elements ICP requires, yet no triple satisfies the ICP. This is tight: ICP is formulable only for $N \geq 5$.*
(3) *$D \not\Rightarrow C$: An extensional 2-pointed magma of size 4 satisfying D has only 2 core elements, so ICP is vacuously unsatisfiable (pigeonhole).*
(4) *$C \not\Rightarrow D$: An extensional 2-pointed magma of size 5 satisfies ICP but violates the classifier dichotomy. This is tight: ICP requires $N \geq 5$.*
(5) *$D \not\Rightarrow S$: An extensional 2-pointed magma of size 4 satisfies the classifier dichotomy but admits no retraction pair. This is tight: D requires $N \geq 4$.*
(6) *$C \not\Rightarrow S$: An extensional 2-pointed magma of size 5 satisfies ICP but admits no retraction pair. This is tight: ICP requires $N \geq 5$.*

Proof. Each direction is proved by exhibiting a concrete Cayley table and verifying the claimed properties by `native_decide` (or `decide` for $N \leq 8$). The tables are:

$S \not\Rightarrow D$ ($N=5$, tight). A 5×5 Cayley table (Appendix A) defining an FRM with $s = r = 2$, the identity on core. Element 3 is a classifier, element 2 is a pure non-classifier, and element 4 is mixed: $4 \cdot 2 = 0$ is absorber-valued while $4 \cdot 3 = 2$ is core-valued, violating the dichotomy. All three dichotomy roles are inhabited, so the failure is non-vacuous (SAT confirms the non-implication is achievable as early as $N=3$, but vacuously: with only 1 core element the dichotomy cannot be formulated). The bound is tight: a classifier, a pure non-classifier, and a mixed element are necessarily pairwise distinct core elements, so a non-vacuous dichotomy failure forces $|S| \geq 5$ — a pigeonhole argument valid for arbitrary binary operations (`no_nonvacuous_dichotomy_failure_at_4`). The earlier $N = 8$ witness (`Countermodel.lean`), found by SAT search over separated role assignments ($s \neq r$, disjoint classifier and mixed elements), is retained; the tight witness shows that collapsing the retraction pair ($s = r$, the identity row) shrinks the counterexample to the minimum. Lean file: `TightWitnesses.lean`, theorem `s_not_implies_d_tight`.

$S \not\Rightarrow C$ ($N = 5$, structural, tight). A 5×5 Cayley table (Appendix A) whose three core elements act on the core as the three transpositions $L_2 = (3\ 4)$, $L_3 = (2\ 3)$, $L_4 = (2\ 4)$. Each L_y is an involution, so every core element is its own retraction pair — S holds three ways over. But the composition of two distinct transpositions moves three points and is never a transposition, so no factorization $L_a = L_c \circ L_b$ over pairwise distinct core elements exists: no triple satisfies the ICP, and the failure is structural, with the core exactly as large as the 3 elements ICP requires. The bound is tight, since ICP is formulable only for $N \geq 5$. The earlier $N=6$ witness with a 4-element core (`E2PM.lean`, `s_not_implies_icp_structural`) is retained. Lean file: `TightWitnesses.lean`, theorem `s_not_implies_icp_structural_tight`.

$D \not\Rightarrow C$ ($N=4$, tight). The minimal S+D model at $N=4$ (`kripke4` in Appendix A) has only 2 core elements. ICP requires 3 pairwise distinct non-absorbers, so it is vacuously unsatisfiable. Lean file: `ICP.lean`, theorem `kripke4_no_icp`. A structural witness exists already at $N = 5$, the minimum size at which ICP is formulable: `TightWitnesses.lean` exhibits both a D-only witness (`d_not_implies_icp_str`
and a full S+D witness of size 5 with no ICP triple (`sd_not_implies_icp_tight`, a `DichotomicRetractMagma` with one classifier and two non-classifiers — by the role-counting argument of the $N = 5$ structure theorem, one classifier makes ICP impossible). The $N = 10$ D+S model in `Countermodels10.lean` is retained: it shows the failure persists with 8 core elements.

$C \not\Rightarrow D$ ($N = 5$, tight). A 5×5 Cayley table satisfying ICP but with no classifier: all three core elements map core to core (no element produces only absorber-valued outputs on core). The bound is tight: ICP requires $N \geq 5$. Lean file: `E2PM.lean`, theorem `h_not_implies_d_tight`.

$D \not\Rightarrow S$ ($N = 4$, tight). A 4×4 Cayley table with the classifier dichotomy (classifier at index 2, non-classifier at index 3) but no retraction pair among the 4 possible assignments from core. The bound is tight: D requires $N \geq 4$. Lean file: `E2PM.lean`, theorem `d_not_implies_s_tight`.

$C \not\Rightarrow S$ ($N = 5$, tight). A 5×5 Cayley table satisfying ICP but admitting no retraction pair. The bound is tight: ICP requires $N \geq 5$. Lean file: `E2PM.lean`, theorem `h_not_implies_s_tight`.

All tables are generated by Z3 SAT solver and independently verified. The Cayley tables are frozen in the repository for reproducibility. ◻

Methodology and minimality. Each counterexample was found by encoding the relevant capability axioms (and the negation of the target capability) as a SAT problem over an $n \times n$ Cayley table, then solving with Z3, incrementing n from 3 upward; each table was frozen and independently verified in Lean. All six directions carry provably tight bounds. D⇏S and D⇏C are tight at $N=4$ (the minimum where D is non-vacuous, resp. where ICP is dimensionally impossible); C⇏S and C⇏D are tight at $N=5$ (the minimum where ICP can be stated); S⇏D is tight at $N=5$ for non-vacuous failure (the classifier, non-classifier, and mixed roles occupy three distinct core slots); and S⇏C is tight at $N=5$ for structural failure (ICP must be formulable). The last two resolve a question left open in an earlier version of this paper: the SAT search producing the original $N=8$ and $N=6$ witnesses ranged over separated role assignments, and overlapping assignments — a collapsed retraction pair $s = r$ whose row doubles as identity or involution — yield strictly smaller witnesses.

Scaling. The independence is not a small-size artifact, and this is a theorem rather than an empirical observation:

Theorem 6.2 (Non-implications and Boolean cube at every size). *For every $n \geq 5$, each of the six non-implications of Theorem 6.1 has a witness of size exactly n, and each of the eight cells of the (S, D, C) Boolean cube (Theorem 6.3) is realized at size exactly n. The cells $\neg S \wedge D \wedge \neg C$ and $\neg S \wedge \neg D \wedge \neg C$ are realized from $n \geq 4$.*

Proof sketch. Seven explicit parametric families of Cayley tables, one per capability profile (the coexistence profile is Theorem 7.3's construction). Each family pads a small witness with three reusable row patterns: identity rows (preserving S), constant-on-core rows (blocking both injective sections and non-trivial factorizations), and transposition rows (preserving S while blocking factorization, since the composition of two distinct transpositions is never a transposition). All properties are verified in Lean by symbolic case analysis uniformly in n — no `decide`. Lean file: `IndependenceAllN.lean`, theorems `independence_all_N` and `boolean_cube_all_N`. □

An earlier version of this claim rested on SAT search, which confirmed satisfiability of all six directions through $N = 15$ (the solver limit); the parametric families supersede that evidence and settle the scaling conjecture.

Interpretation. The independence result shows that S, D, and C are genuinely three things, not one. An algebra can encode and decode without classifying (S without D). An algebra can classify without encoding (D without S). An algebra can internally compose without classifying or encoding (C without S or D). No capability algebraically forces any other. In particular, the $C \not\Rightarrow D$ direction shows that an algebra can internalize a non-trivial composition pattern without satisfying the classifier dichotomy: internal composition does not force organizational separation.

6.1 Joint irredundance: the Boolean cube

Pairwise independence shows no capability implies any other. A stronger form of irredundance asks whether some capability is a *Boolean function* of the other two: if S were equivalent to $D \wedge \neg C$, say, the triple (S, D, C) would collapse to the pair (D, C). We rule this out by exhibiting a witness in each of the eight cells of the (S, D, C) Boolean cube.

Theorem 6.3 (Joint irredundance). *Each of the eight Boolean combinations of (S, D, C) is realized by a Lean-verified finite extensional 2-pointed magma.*

Cell	Profile	N	Witness
1	$S \wedge D \wedge C$	5	`witness5`
2	$S \wedge D \wedge \neg C$	4	`kripke4`
3	$S \wedge \neg D \wedge C$	5	`famCm` (`IndependenceAllN`)
4	$S \wedge \neg D \wedge \neg C$	3	`frm3` (new)
5	$\neg S \wedge D \wedge C$	5	`cell5` (new)
6	$\neg S \wedge D \wedge \neg C$	4	`dNoS4`
7	$\neg S \wedge \neg D \wedge C$	5	`hNoD5`
8	$\neg S \wedge \neg D \wedge \neg C$	3	`cell8` (new)

Four cells are populated by witnesses that already appear in the pairwise-independence proof, after adding the missing capability annotations; cell 3 is realized tightly at $N=5$ by the parametric family `famCm` (previously only an $N=10$ witness, `hNotD`, was known). Three required fresh constructions: cell 4 and cell 8 are minimal ($N = 3$) FRM and E2PM witnessing "S alone" and "no capability" respectively; cell 5 (D and C without S) is the cube's most structurally constrained corner, an $N = 5$ magma with two classifiers, one core-preserving non-classifier serving as the ICP's b, and no retraction pair. All verifications are machine-checked by `decide` or `native_decide` in `BooleanCube.lean`.

By Theorem 6.2, every cell is in fact realized at *every* size $n \geq 5$ (cells 6 and 8 from $n \geq 4$), so joint irredundance is not a small-size artifact either. The minimal cell sizes reveal an asymmetry: every cell containing C requires $N \geq 5$ — and all four C-cells are realized exactly at $N = 5$, which is tight — while cells with only S and/or D are realized at $N = 3$ or 4. C carries strictly more structural weight than S or D, consistent with the row-scarcity argument in Section 10: ICP's factorization demand $a \cdot x = c \cdot (b \cdot x)$ constrains three rows simultaneously and requires three pairwise-distinct core elements, which S and D do not.

Consequence. If any capability were Boolean-equivalent to a combination of the other two, some cell would be empty. All cells are inhabited: no reduction of three predicates to two exists. S, D, and C therefore carry mutually irreducible information — jointly irredundant, not merely pairwise independent.

7 COEXISTENCE

Theorem 7.1 (No ICP at $N=4$). *No extensional 2-pointed magma of size 4 satisfies the ICP. The core $\{2, 3\}$ has only 2 elements, but ICP requires 3 pairwise distinct non-absorbers. Lean:* `Witness5.lean`, `no_icp_at_4`.

Theorem 7.2 (Coexistence Witness). *The three capabilities S, D, and C are simultaneously satisfiable. A Lean-verified witness exists at $N=5$, which is optimal by Theorem 7.1.*

Proof. A concrete 5×5 Cayley table is exhibited with $s = r = 2$ (identity on core), classifiers $\tau \in \{3, 4\}$, and ICP witnessed by $a = 3, b = 2, c = 4$. The retraction pair collapses to a single element ($s = r$) which doubles as the ICP's core-preserving map; since b is the identity, the factorization $a \cdot x = c \cdot (b \cdot x)$ reduces to $a \cdot x = c \cdot x$ on core, yielding two classifiers that agree on core but distinguished by their absorber columns. The lower bound is pigeonhole: at $N=4$, the core $\{2, 3\}$ has only 2 elements, but ICP requires 3 pairwise distinct non-absorbers (`no_icp_at_4`). Lean file: `Witness5.lean`, theorem `sdh_witness_5`. A 6×6 witness with $s \neq r$ is also verified (`Witness6.lean`). □

Theorem 7.3 (Scaling: S+D+C exists at all $N \geq 5$). *For every integer $N \geq 5$, there exists an extensional 2-pointed magma on $\mathrm{Fin}(N)$ satisfying S, D, and C. Lean:* `WitnessAllN.lean`*, theorem* `sdh_witness_all_N`.

Proof (constructive). For $N \geq 5$, define $T_N : \mathrm{Fin}(N)\times\mathrm{Fin}(N) \to \mathrm{Fin}(N)$ by:

- $T_N(0,x) = 0$ for all x.
- $T_N(1,x) = 1$ for all x.
- $T_N(2,x) = x$ for all x.
- $T_N(3,x) = 1$ if $x \in \{3,4\}$, else $T_N(3,x) = 0$.
- $T_N(4,x) = 1$ if $x \in \{1,3,4\}$, else $T_N(4,x) = 0$.
- For $k \in \{5,\ldots,N-1\}$: $T_N(k,2) = k$, $T_N(k,k) = 2$, and $T_N(k,x) = x$ for all $x \notin \{2,k\}$.

E2PM axioms. Rows 0 and 1 are constant absorbers. For $k \in \{2,3,4\}$ row k takes at least two values, so no row beyond rows $0,1$ is constant. Extensionality: rows $0,1$ differ from each other and from all higher rows at column 0 (row 0 has 0, row 1 has 1, row $k \geq 2$ has 0). Row 2 differs from rows $3,4$ at column 2 (row 2 has 2, rows $3,4$ have 0). Rows 3 and 4 differ at column 1. For $k,k' \in \{5,\ldots,N-1\}$ with $k \neq k'$: rows k and k' differ at column k (row k has 2, row k' has k).

S. Take $s = r = 2$. Anchoring $T_N(2,0) = 0$ holds. The retraction equations $T_N(2,T_N(2,x)) = T_N(2,x) = x$ on $\mathrm{core}(S)$ hold trivially since row 2 is identity.

D. Rows 3 and 4 have values in $\{0,1\}$ everywhere, so they are full classifiers. Rows 2 and k for $k \geq 5$ map $\mathrm{core}(S)$ into $\mathrm{core}(S)$ (verified by inspection), so they are non-classifiers. The dichotomy holds. Take $\tau = 3$ as the full classifier; the non-classifier 2 exists.

C. The triple $(3,2,4)$ is a C-triple: the three indices are pairwise distinct in $\mathrm{core}(S)$; row 2 is core-preserving; for $x \in \mathrm{core}(S)$, $T_N(3,x) = T_N(4,T_N(2,x))$ holds because row 2 is identity, so $T_N(3,x) = T_N(4,T_N(2,x)) = T_N(4,x)$ reduces to row equality on core, which holds by direct comparison (both rows are constant 0 on core $\setminus \{3,4\}$ and constant 1 on $\{3,4\}$); $T_N(3,\cdot)|_{\mathrm{core}(S)}$ takes both values 0 and 1 (at $x = 2$ and $x = 3$ respectively), so non-triviality holds. □

Proposition 7.4 (Separated-Role Witness). *There is a witness at $N = 10$ in which the principal S+D+C roles (2 absorbers, 2 retraction pair, 1 classifier, 3 ICP witnesses) occupy 8 distinct elements.*

Proof. A concrete 10×10 Cayley table (Appendix A) satisfies all three capabilities with distinct role assignments. Lean file: `Witness10.lean`. A witness with $s \neq r$ (non-degenerate retraction) is exhibited at $N = 6$ (Appendix A). □

Minimum cardinality. $N = 5$ is the optimal minimum for S+D+C coexistence (Lean-verified; `Witness5.lean`). The witness uses $s = r$ (identity on core), two classifiers, and one non-classifier: the retraction pair member doubles as the ICP's core-preserving element. At $N = 4$ the core has only 2 elements but ICP requires 3 pairwise distinct non-absorbers, so $N = 4$ is impossible (`no_icp_at_4`).

The $N = 5$ witness reveals how thin the three properties can become simultaneously. The retraction pair is the identity on core ($s = r$, no actual encoding work), and the ICP factorization $a \cdot x = c \cdot (b \cdot x)$ collapses to $a \cdot x = c \cdot x$ (row equality on core between two classifiers differing only on absorber columns). The three capabilities are formally present but barely interacting. The $N = 6$ witness with $s \neq r$ (`Witness6.lean`) is the more informative example: the retraction pair does genuine encoding, the ICP factorization is non-degenerate, and the roles do not trivially overlap. We retain the general definitions (allowing $s = r$) since the algebraic results (independence, injectivity, no-associativity) hold regardless.

8 CONNECTING AXIOMS: SORTING, INTROSPECTION, AND THE CANONICAL MODEL

The independence and irredundance theorems say that S, D, and C never force one another: any system in which the capabilities cohere does so by *additional* axioms. This section identifies and studies the first such connecting axiom.

Under D, every core element is a classifier (boolean on core) or a non-classifier (core-valued on core), so the core carries a two-valued *class map*. The dichotomy already makes classifier rows compositional for free: a classifier's core outputs land in the absorbers, whose status is fixed. What D leaves entirely open is the non-classifier rows: for $n \in N$ and core x, the output $n \cdot x$ is core, but whether it is a classifier or a non-classifier may depend arbitrarily on both n and x.

Definition 8.1 (Sorting). An extensional 2-pointed magma satisfying D is *sorted* when the class of $n \cdot x$ depends only on the class of x, uniformly in the non-classifier n and the representative x. Equivalently: the $Z/C/N$ decomposition is compositional on the core, i.e. the class map is a homomorphism-like abstraction of the operation — which is what a type discipline is, algebraically. Lean: `Sorting.lean`, `Sorted`.

Sorting is "the other half" of a type system: D sorts the classifier rows automatically, sorting sorts the rest. Three results locate the axiom precisely.

Theorem 8.2 (Sorted involution). *In a sorted dichotomic magma with a classifier, a non-classifier, and a retraction pair (whose members are non-classifiers, automatic by the placement theorem; only the one-sided equation is used), the induced class-action of the non-classifiers is an involution: either every non-classifier preserves both classes, or every non-classifier swaps them. Lean:* `sorted_involution` *(algebraic, uniform in N).*

Proof sketch. Sorting makes all non-classifiers share one class-action $f : \{C,N\} \to \{C,N\}$. The round trips $r \cdot (s \cdot \tau) = \tau$ and $r \cdot (s \cdot n_0) = n_0$ through the section force $f(f(t)) = t$ on both classes, and an involution on a two-element set is the identity or the swap. □

This is the first genuine interaction theorem between the capabilities: S constrains the shape of sorted D, cutting the four conceivable class-tables down to two "sorted worlds." Neither is degenerate, and the axiom itself is genuinely new:

Proposition 8.3 (Independence and inhabitation). *(i) The canonical $N = 5$ coexistence witness is sorted, in the class-preserving world (*`witness5_sorted`*,* `witness5_class_preserving`*). (ii) Sorting is not implied by S+D+C, even at $N = 5$ where the role assignment is otherwise completely forced: an $N = 5$ S+D+C magma exists* *whose unique non-classifier acts as the transposition* $(\tau_1\ b)$*, sending the two classifiers to different classes (*`sorting_independent`*). (iii) The class-swapping world is inhabited at* $N=6$*: an S+D+C magma with a non-degenerate retraction pair* $(s \neq r)$ *whose section exchanges the classifier block with the non-classifier block (*`swap_world_inhabited`, `swap6_swaps`*). The swap world cannot exist at* $N=5$*: the section must inject the classifiers into the non-classifiers, so* $|N| \geq |C|$*, while the structure theorem forces* $|C| = 2$*,* $|N| = 1$*. Cayley tables in Appendix A.*

The classification is complete: all four conceivable class-tables are realized, and S permits exactly the involutive two.

Proposition 8.4 (The four class-tables). *Each of the four class-actions* $f : \{C, N\} \to \{C, N\}$ *is realized by a sorted dichotomic magma: the identity by* `witness5`*, the swap by* `swap6`*, the constant-C table by the tight* $D \not\Rightarrow S$ *witness* `dNoS4` *(its unique non-classifier sends everything to the classifier), and the constant-N table by a new* $N=4$ *witness (*`constN4`*, Appendix A). The two constant worlds* force $\neg S$*: by the involution theorem, a sorted magma with a retraction pair lies in the identity or swap world, so any constant world excludes retraction pairs outright (*`constC_blocks_retraction`, `constN_blocks_retraction` *— corollaries of Theorem 8.2; both constant witnesses are also directly verified to admit no retraction pair). S is thus exactly the axiom that separates the involutive class-tables from the constant ones.*

Theorem 8.5 (Swap balance). *In the class-swapping world, a retraction pair forces* $|C| = |N|$*: the section injects each class into the other, and injectivity in both directions balances the counts (*`swap_balance`*; only the one-sided retraction equation is used). Under D the two classes partition the core, so the core has even size (*`swap_even_core`*). Consequently, within S+D+C the swap world can exist only at even* $N \geq 6$*, and* `swap6` *realizes its minimum; at every odd size the sorted S+D+C landscape is confined to the class-preserving world.*

The two S-compatible worlds are then fully separated by size:

Proposition 8.6 (Size spectrum of the sorted worlds). *The class-preserving world exists at every* $N \geq 5$*: the canonical scaling family of Theorem 7.3 is itself sorted and class-preserving (*`witnessAllN_sorted`, `witnessAllN_class_preserving` *— algebraic, uniform in* N*, via the bridge lemmas* `sorted_of_preserving` / `sorted_of_swapping` *showing that either class-action implies sortedness). The class-swapping world exists only at even core sizes (Theorem 8.5), with minimum* $N=6$*.*

The arc is progressive rigidification: D alone sorts the classifier rows; the sorting axiom makes the class map compositional; adding S cuts the sorted worlds from four to two (and parity can cut them to one); at $N=5$, cardinality forces the class-preserving world outright. Under the computational reading — preserving as stratified typing, swapping as quotation — the typed world is available at every size, while the quoting world demands an exact balance between judges and data.

8.1 Homoiconic introspection

The next connecting axiom after sorting is *sort-introspection*: a classifier κ that internally decides the C/N partition itself ($\kappa \cdot y = z_1$ on classifiers, z_2 on non-classifiers — `data?`/`judge?` in Lisp terms). A SAT sweep over sorted S+D+C magmas in both worlds ($N = 6, 8, 10, 16$) shows that introspection *determines the quotation law*, and the determination is Lean-proved (`Homoiconic.lean`):

Theorem 8.7 (Introspection fixes the quotation law). *Let a sorted dichotomic magma carry a retraction pair and a sort-introspective classifier* κ*. In the class-swapping world,* κ negates *under quotation:* $\kappa \cdot (s \cdot x) \neq \kappa \cdot x$ *for every core* x *— "x is a judge iff* $s \cdot x$ *is data" — and a quote-transparent introspective classifier is impossible (*`introspection_negates_c`*; UNSAT at* $N = 6, 8, 10, 16$*). Dually, in the class-preserving world* κ *is quote-transparent, and a quote-negating introspective classifier is impossible (*`introspection_transparent_of_preserving`*).*

A homoiconic system therefore does not choose its introspection law: typing forces transparency, quoting forces negation. The minimal magma carrying the full stack — S, D, C, sorted swap world, introspection, negation, judge-closure — exists at $N = 6$ and reads as a complete Lisp kernel (Appendix A): two halt states, `quote`, `eval`, `data?`, `judge?`, with the single internal composition being the homoiconicity law itself, `judge?` = `data?`∘`quote` (`canonical_kernel`, `kernel6_icp_through_quote`). In this kernel quotation has order 4 on the core: $\texttt{quote}^2$ is a non-trivial involution and $\texttt{quote}^4 = \mathrm{id}$ (`kernel6_quote_order_four`). One further SAT observation: in the preserving world, universal quote-negation is satisfiable exactly when the block size $(N-2)/2$ is even (UNSAT at $N=8$, SAT at $N = 6, 10$) — an odd block admits no fixed-point-free alternating 2-coloring of the quote orbits — a third parity law, recorded here as a SAT result with a hand proof.

8.2 The canonical eight-element model

Extending the kernel by one free dual pair under a full law set — faithfulness of the free operator, hygiene (commutation with quote and eval, involutivity), distinct action, and judge-closure — yields a design space of exactly 228 core tables at $N=8$ (SAT enumeration, `scripts/n8_free_pair_search.py`). Its lexicographically minimal member is frozen and verified in `ArtifactN8.lean` (`canonical_artifact_N8`; Appendix A): eight instructions reading as halt-true, halt-false, `quote`, `eval`, `shift`, `data?`, `judge?`, `shift?`. Three emergent regularities appear without being imposed: the duality pairing documents the instruction set (each operator's code *is* its judge: `quote`·`quote` = `data?`, `quote`·`eval` = `judge?`, `quote`·`shift` = `shift?`); the unconstrained judge slot resolves into the recognizer of exactly {`shift`, `quote` · `shift`}; and quotation is an involution.

Two further impossibility observations shape the model's architecture: a faithful constructor cannot have a non-trivial internal recognizer of its image (injectivity forces surjectivity on a finite core), and no element of any swap-world magma can realize $\texttt{quote}^2$ (block-preserving, hence expressible by no row) — so data structure and derived actions necessarily live outside the algebra, in the external driver that K-infinity (Theorem 3.14) already requires.

8.3 The derivation ledger: forced, free, and chosen

The law set that carves out the 168-model space contains candidate axioms of unequal status. Three groups of theorems, all algebraic and uniform in N, reduce the set: the first derives the model's frame (world and size) from a single semantic requirement, the second shows the eval side of the law set is redundant, and the third reduces the reversibility and closure laws to tie-breaks. These are the provenance results that the companion machine-level write-up imports when it treats the model as a found object.

Theorem 8.8 (Observable quotation forces the frame). *Let M be a sorted dichotomic magma with a retraction pair (s, r) (members non-classifiers, automatic by the placement theorem; only the one-sided equation is used) and a sort-introspective classifier κ that* observes quotation*: $\kappa \cdot (s \cdot x) \neq \kappa \cdot x$ for every core x. Then M lies in the class-swapping world (*`swap_of_observable_quotation`*). If M moreover carries a faithful core operator γ (injective on core) with s, r, γ pairwise distinct, then $|S| \geq 8$, and the canonical model attains the bound, so it is sharp (*`stack_a_frame_min`, `stack_a_frame_attained`*;* `StackAForced.lean`*).*

Proof sketch. By the involution theorem (Theorem 8.2) the world is preserving or swapping, and the determination theorem (Theorem 8.7) makes preserving worlds quote-transparent, contradicting observability — so the world swaps. A faithful operator cannot sit on the classifier side once the core has three distinct elements (two absorber values cannot receive three core elements injectively), so s, r, γ are three distinct non-classifiers; swap balance (Theorem 8.5) forces at least three classifiers to match; the two absorbers complete the count: $3 + 3 + 2 = 8$. □

Theorem 8.9 (The eval side of the law set is free). *Let (s, r) be a mutual retraction pair on core, both core-preserving. (i) Any core-preserving operator γ that commutes with s on core commutes with r on core, by conjugation through the retraction (*`eval_comm_of_quote_comm`*). (ii) If the quote-precomposition $t \circ s$ of every judge t is realized by a judge, then so is the eval-precomposition $t \circ r$ of every judge: iterating the closure hypothesis realizes $t \circ s^m$ for every m, pigeonhole on the finite carrier yields $t \equiv t \circ s^d$ on core for some $d \geq 1$, and then $t \circ r \equiv t \circ s^{d-1}$, realized by the sequence (*`eval_closure_of_quote_closure`*;* `EvalSideFree.lean`*). Consequently two eval-side laws of the canonical model's law set are redundant: shift-commutes-with-eval follows from shift-commutes-with-quote, and eval-side judge-closure from quote-side. Deleting the former is empirically re-verified to leave the model space unchanged (*`scripts/n8_enumerate_lexmin.py`*).*

Theorem 8.10 (Reversibility, parity, and orbit closure are free). *(i) Every faithful core-preserving operator γ has finite order on the core: some $d \geq 1$ satisfies $\gamma^d = \mathrm{id}$ on core (*`faithful_finite_order`*); in particular iterated quotation always cycles (*`quote_finite_order`*) — a hygienic renaming operator is automatically undoable, by iterating itself. (ii) In the class-swapping world that order is necessarily* even*: odd powers swap the classifier blocks, so no odd power is the identity (*`swap_even_order`*). The canonical model's involution law is therefore exactly the choice of the* minimal *order — a tie-break in the same family as lex-min, not an axiom of substance. (iii) If quotation has core order d and the d actions $\kappa \circ s^m$ ($m < d$) of the introspector's quote-orbit are realized by judges, that family is closed under precomposition with both quote and eval — modular arithmetic on the orbit (*`orbit_quote_closure`, `orbit_eval_closure`*;* `QuoteOrbit.lean`*). The residual content of the judge-closure law concerns only judges* outside *the introspector's orbit — at $N = 8$ with involutive quote, exactly one free judge, which lex-minimality resolves into the recognizer* `shift?`.

The ledger. After these reductions the canonical model's law set separates cleanly. *Derived*: the class-swapping world, the carrier size ($N \geq 8$, sharp), the entire eval side, reversibility of the hygiene operator, the evenness of its order, and closure of the realized quote-orbit. *Chosen*: quote-commutation (the *definition* of hygienic), order = minimum (a tie-break), orbit-realization plus closure for judges outside the orbit, shift's action-distinctness, and the lexicographic tie-break itself. The residue of arbitrariness is finite, named, and public; everything that shrank is now labeled theorem rather than choice. (Axiom footprint: unlike the `decide`-checked finite witnesses, Theorems 8.8–8.10 go through Mathlib's finiteness machinery and so additionally depend on `Classical.choice` and `Quot.sound`.)

Computational reading. The swap world first appearing at $N = 6$ is one more face of the $N = 5 \rightarrow N = 6$ phase transition. In the class-swapping world the section literally encodes judges as data and data as judges — the algebraic shadow of quotation exchanging program and value — while the class-preserving world is the shadow of a stratified (typed) discipline in which encoding respects sorts. That the minimal connecting axiom already bifurcates into exactly these two shapes is, we believe, the beginning of an answer to what a type system is in this setting: sorting is the axiom, and the sorted worlds are its two consistent implementations. The machine-level consequences — two-level driver architecture, heap tags, driver-side user quotation — are developed in the companion write-up, which imports the canonical model of Section 8.2 together with the ledger above as its found object.

9 RELATED WORK

Partial and total combinatory algebras. PCAs [10, 11] and TCAs [12] provide the standard algebraic framework for realizability and computability. PCA theory has a rich structural theory: morphisms between PCAs, the lattice of PCA quotients, and totalizability [13] all involve detailed analysis of the application operation. PCA theory does study which elements realize which functions. However, this analysis concerns representability and inter-PCA relationships, not the *partition* of elements into behavioral types by their row structure. This question requires a finite total operation table to formulate; it does not arise in the PCA/TCA setting, where application is partial or the carrier is infinite. Bethke and Klop [13] proved that some PCAs cannot be extended to total ones, a structural result about the relationship between partiality and totality. The K-infinity theorem (our Theorem 3.14) shows that any nontrivial total magma with a **k**-combinator must be infinite — in particular, all combinatory algebras — explaining why finite algebras fall outside this framework and why our setting requires a different combinator basis.

Turing categories. Cockett and Hofstra [14] define a Turing category as a cartesian restriction category with a Turing object A and a Turing morphism $\bullet : A \times A \rightarrow A$ that universally simulates all morphisms in the category. The Structure Theorem establishes that PCAs and Turing categories are two views of the same concept. Turing objects have cardinality "one or infinite" [14], ruling out nontrivial finite Turing objects. The Turing morphism is described as "the heart of the whole structure" but its algebraic properties (row structure, element classification, associativity constraints) are not studied. Where Turing categories distribute computational roles across distinct objects (programs, data, evaluators), our finite algebras develop the same roles internally as emergent behavioral regions within a single carrier set: absorbers encode failure, classifiers provide absorber-valued judgment, and the retraction pair provides encoding.

Self-interpretation. Mogensen [1] constructs self-interpreters for the λ-calculus. Brown and Palsberg [2] show that self-interpretation is possible in strongly-normalizing typed languages, contradicting earlier impossibility results. These works study self-interpretation (our S) but do not address the independence of S from D or C.

Independence in finite algebra. Independence and non-implication results between equational properties have a long history in universal algebra, from Birkhoff's characterization of equational classes [6] to Mal'cev conditions characterizing implications between properties of varieties. Our results operate at a different level: we separate *operational* properties (retraction pairs, classifier partitions, factorizations in the left-regular representation) rather than equational identities, and our base class, the finite extensional 2-pointed magmas, is not a variety in the Birkhoff sense (extensionality is a quasi-identity, not an equation). To our knowledge, the pairwise independence of retraction pairs, classifier dichotomies, and factorization properties in finite magmas has not been previously established. The SAT-find-then-Lean-verify methodology is related to the use of automated tools such as Mace4 [8] for counterexample search in algebra, though we use Z3 for constraint solving and Lean 4 for independent formal verification.

Finite model theory and machine-checked proofs. Independence results via finite counterexamples are standard in finite model theory [5]. Each Cayley table is frozen in the Lean source and verified by the kernel, eliminating transcription errors. The sizes involved ($N = 5$ through $N = 10$) are small enough for `decide`/`native_decide` but large enough that hand-verification of all conditions would be error-prone. The universal algebraic theorems (no-associativity, decomposition invariance, ICP equivalence) are proved by pure equational reasoning without `decide`, and hold for all finite sizes.

Cellular automata and applicative calculus. The structures studied here sit at an intersection of two computational-substrate traditions that have otherwise remained separate. Cellular automata (Rule 110, Conway's Game of Life, and the broader tradition) establish universality from a finite local rule by iterated unfolding; the rule itself is small and fixed, while computation emerges from temporal-spatial extent. Applicative calculi (combinatory logic, the λ-calculus) take the binary application operation as primitive and use extensionality to individuate functions by behavior. Finite extensional 2-pointed magmas inherit the finite-local-rule nature of cellular automata (the Cayley table is the rule; computation unfolds as a term tree) together with the applicative-and-extensional structure of combinatory calculi (the operation is $a \cdot b$, asymmetric between operator and operand, with rows individuating elements). They differ from each tradition in specific ways: from cellular automata, they drop the spatial grid, replacing it with term-tree unfolding; from combinatory calculi, they drop abstraction, fixing the entire function inventory as the finite carrier. Neither tradition has systematically studied the resulting hybrid — finite applicative substrates with extensional individuation — and in particular the capability-decomposition questions (what can be internalized as element behavior modulo the operation?) answered by this paper do not arise naturally in either parent.

10 DISCUSSION

The classifier dichotomy is organizational. D organizes elements into coherent categories: classifiers that test and non-classifiers that transform. The $C \not\Rightarrow D$ result shows that an algebra can internalize composition without this organization. D contributes semantic structure rather than compositional behavior.

The role of extensionality. Extensionality (distinct elements have distinct rows) is load-bearing throughout. Without it, the no-associativity proof fails: Step 4 derives $\tau = v$ from identical rows. Without it, any constant-row element trivially satisfies both classifier and non-classifier conditions, making the three-category decomposition degenerate. Extensionality is not merely a convenience but the condition that makes element-level classification meaningful. An element's identity *is* its row in the Cayley table; extensionality formalizes this.

Why these three properties? Section 2 motivated S, D, and C as the three fundamental structural questions about the row-map family $\{L_a^{\mathrm{core}}\}$ — splitting, codomain purity, and factorization — each minimal in its own sense: the standard section-retraction notion, the coarsest invariant partition, and the weakest non-degenerate factorization (weakening either ICP guard condition makes every retraction-equipped magma satisfy it trivially; Lean-verified separations in `ICP.lean`). The independence and joint-irredundance results (Theorems 6.1, 6.3) confirm these are three questions jointly irredundant, not reducible to two. The precise categorical status of D and C is discussed in the categorical connections paragraph below; whether either corresponds to a known universal property remains open.

(S, D, C) as a structurally distinguished slice. A Z3 canonicality probe (`scripts/canonicality/probe_axioms.py`) tested thirteen candidate operational axioms on E2PM at $N \in \{5, 6\}$ via pairwise directional implication queries against S, D, C; ten are pairwise independent of (S, D, C) in both directions at both sizes, three collapse (S subsumes one-sided retraction and left-cancellation; both D and C imply the common substructure P via Theorem 3.3). So (S, D, C) is one slice of a structurally fertile axiom landscape, not an exhaustive axiomatisation. What distinguishes this particular slice is that it is structurally selectable: at $N = 5$ the canonical S+D+C witness (Remark 10, Appendix A) is fully determined by the topos-theoretic indicator reading of D plus the requirement that the magma's automorphism be internalised in its operation. The other ten probe axioms have no such joint structural characterisation.

Categorical connections and their limits. The core core(S) is a retract of the carrier via (s, r) in the standard sense of [3]; the three-category decomposition defines a functor from **E2PM** to three-block set partitions (Theorems 4.2, 4.4); D is a codomain-purity condition on the row maps $\{L_a^{\mathrm{core}}\}$. The categorical analogy stops at composition: the row maps are *not* closed under composition (the no-associativity theorem predicts this — associativity would be a prerequisite for the row family to form a monoid), so standard categorical constructions (internal hom, enrichment, Turing morphism) cannot be stated internally. The composition-closure failure is a consequence of the same obstruction that motivates the setting.

Scope and limitations. Whether the three-capability separation appears in other settings (λ-calculus, typed systems, infinite algebras) is the primary open question. Theorem 3.10 constrains the target: any such translation must work without associativity or identity. Each capability admits multiple axiom forms; the independence result holds for all variants. The independence also implies that any system combining all three capabilities must introduce additional axioms beyond S, D, and C to connect them; Section 8 takes the first step, identifying sorting (class-compositionality) as such an axiom, proving it independent of S+D+C even at the minimal coexistence size, and classifying the sorted worlds into exactly two shapes via the involution theorem.

11 CONCLUSION

Four obstructions — K-infinity, no-associativity, asymmetry, and single-absorber collapse — jointly narrow the candidate class of finite magmas that can carry the operational capabilities of splitting, dichotomy, and composition down to the setting of this paper: finite, non-associative, non-commutative, extensional 2-pointed magmas. Within it, we have identified three natural algebraic properties, internal splitting (S), the classifier dichotomy (D), and the Internal Composition Property (C), and proved they are pairwise independent. The six non-implications are machine-checked in Lean 4, each at its provably minimal size. Joint irredundance strengthens this: all eight Boolean combinations of (S, D, C) are realized (Theorem 6.3), so no capability is a Boolean function of the other two; parametric families realize every non-implication and every Boolean profile at every admissible carrier size (Theorem 6.2), so neither result is a small-size artifact. The optimal coexistence witness has $N = 5$ (Theorem 7.2). The three-category decomposition induced by D is a functorial invariant (Theorem 4.2); most principal witnesses are role-rigid, with the canonical $N = 5$ S+D+C witness as the deliberate exception (Proposition 4.6, Remark 10); at $N = 5$ a structure theorem pins retraction, classifier, and C-triple onto the three core slots and rules out strong S altogether (Theorem 4.8), the mirror-row theorem (Theorem 4.11) bounds $|\mathrm{Aut}(M)| \leq 2$, while at $N \geq 6$ strong S returns and rigidity fails (Proposition 4.12); and the ICP is logically equivalent to Compose+Inert (Theorem 5.1). Beyond independence, sorting begins the connecting-axiom programme (Section 8), and the canonical eight-element model is derived rather than designed: its frame is forced by observable quotation (Theorem 8.8), the eval side of its law set is free (Theorem 8.9), and its reversibility, parity, and orbit-closure laws reduce to tie-breaks (Theorem 8.10). Lean-formalized results — independence counterexamples, joint-irredundance witnesses, decomposition invariance, the ICP equivalence, the all-N scaling witnesses (Theorems 7.3 and 6.2), the tight counterexamples (`TightWitnesses.lean`), C-triple type-coherence (Theorem 3.12), the $N = 5$ structure theorem (Theorem 4.8), the mirror-row theorem (Theorem 4.11), and the frame-derivation and law-reduction theorems (Theorems 8.8–8.10) — compile with zero `sorry`. The remaining universal-N algebraic results of Section 3 not yet transcribed to Lean (most notably the P-witness identity of Theorem 3.2 and the classifier-count bound of Theorem 3.13) are proved by hand and supported by Z3 UNSAT certificates.

Open problems.

(1) *Cross-formalism universality*: Does the three-capability separation appear in the λ-calculus, typed settings, or infinite algebras? The no-associativity obstruction (Theorem 3.10) constrains the target setting.
(2) *Categorical characterization of C*: The ICP is the weakest non-degenerate factorization property of the left-regular representation (weakening either pairwise-distinctness or non-triviality makes it vacuous). Is it equivalent to a standard universal property (e.g., existence of a partial internal hom)?
(3) *Categorical characterization of D*: D is the unique coarsest non-trivial isomorphism-invariant partition of core (Corollary 4.3), isolating the two-valued aspect of a subobject classifier without the universality aspect. The left-multiplication maps lack composition closure, precluding standard categorical formulations. Does D correspond to a known concept in non-associative categorical algebra, or does it constitute a new structural property?
(4) *Rigidity propagation beyond $N = 5$*: Theorem 4.8 establishes a rigid role assignment at $N = 5$ (and in particular rules out strong S), but rigidity fails at $N = 6, 7, 8$ under strong S (Proposition 4.12). Several natural single-axiom strengthenings (classifier-nontriviality on core, asymmetric anchoring of the section, anchor monotonicity, strict anchor asymmetry) were each tested and each Z3-satisfiable alongside a non-trivial automorphism at some $N \in \{6, 7, 8\}$ (scripts and witness JSONs under `scripts/`). Characterizing the minimum axiom system that uniformly forces role-rigidity at $N \geq 5$ — and whether it admits a bounded finite specification — is open.

Artifact. All Lean files, counterexample tables, and SAT reproduction scripts are available at https://github.com/stefanopalmieri/finite-magma-independence.

REFERENCES

[1] Torben Mogensen. Efficient self-interpretation in lambda calculus. *Journal of Functional Programming*, 2(3):279–291, 1992.
[2] Matt Brown and Jens Palsberg. Breaking through the normalization barrier: A self-interpreter for F-omega. In *POPL*, pages 5–17, 2016.
[3] Saunders Mac Lane. *Categories for the Working Mathematician*. Springer, 1971.
[4] Peter T. Johnstone. *Sketches of an Elephant: A Topos Theory Compendium*, vol. 1. Oxford University Press, 2002.
[5] Leonid Libkin. *Elements of Finite Model Theory*. Springer, 2004.
[6] Stanley Burris and H. P. Sankappanavar. *A Course in Universal Algebra*. Springer, 1981.
[7] Dana Scott. Continuous lattices. In *Toposes, Algebraic Geometry and Logic*, vol. 274 of *Lecture Notes in Mathematics*, pages 97–136. Springer, 1972.
[8] William McCune. *Prover9 and Mace4*. https://www.cs.unm.edu/~mccune/prover9/, 2005–2010.
[9] Leonardo de Moura and Sebastian Ullrich. The Lean 4 theorem prover and programming language. In *CADE*, pages 625–635, 2021.

[10] Jaap van Oosten. *Realizability: An Introduction to its Categorical Side.* Studies in Logic and the Foundations of Mathematics 152, Elsevier, 2008.
[11] John Longley and Dag Normann. *Higher-Order Computability.* Theory and Applications of Computability, Springer, 2015.
[12] Ingemarie Bethke. *Notes on Partial Combinatory Algebras.* PhD thesis, Universiteit van Amsterdam, 1988.
[13] Ingemarie Bethke and Jan Willem Klop. Collapsing partial combinatory algebras. In *Higher-Order Algebra, Logic, and Term Rewriting (HOA 1995)*, LNCS 1074, pages 16–48, Springer, 1996. DOI: 10.1007/3-540-61254-8_18.
[14] J. Robin B. Cockett and Pieter Hofstra. Introduction to Turing categories. *Annals of Pure and Applied Logic*, 156(2–3):183–209, 2008. DOI: 10.1016/j.apal.2008.04.005.

A CAYLEY TABLES

All tables are extracted from the Lean source files and verified by `lake build`. Element i's row is row i; entry (i, j) is $i \cdot j$.

$N=4$: Minimal S+D witness ($s = r$). Roles: $0 = z_1$, $1 = z_2$, $2 = \tau$, $3 = s = r$. Lean: `Dichotomic.lean`, `kripke4`.

·	0	1	2	3
0	0	0	0	0
1	1	1	1	1
2	0	1	0	1
3	0	0	2	3

$N=5$: Minimal S+D witness ($s \neq r$). Roles: $0 = z_1$, $1 = z_2$, $2 = s$, $3 = r$, $4 = \tau$. Lean: `Dichotomic.lean`, `kripke5`.

·	0	1	2	3	4
0	0	0	0	0	0
1	1	1	1	1	1
2	1	0	3	4	2
3	0	2	4	2	3
4	0	1	1	0	0

$N=5$: Canonical S+D+C Coexistence ($s = r$, minimal coexistence witness). Roles: $0 = z_1$ (FALSE), $1 = z_2$ (TRUE), $2 = \tau_1$ (classifier, indicator of $\{\tau_1, z_2\}$, ICP a), $3 = \tau_2$ (classifier, indicator of $\{\tau_2, z_2\}$, ICP c), $4 = g = s = r$ (non-classifier acting on core as the transposition $(\tau_1\ \tau_2)$, ICP b). Lean: `Witness5.lean`.

·	0	1	2	3	4
0	0	0	0	0	0
1	1	1	1	1	1
2	0	1	1	0	0
3	0	1	0	1	0
4	0	1	3	2	4

$N = 6$: S+D+C Coexistence (role overlap). Roles: $0 = z_1$, $1 = z_2$, $2 = s$ (also ICP a), $3 = r$ (also ICP b), $4 = \tau$, $5 =$ ICP c. Lean: `Witness6.lean`.

·	0	1	2	3	4	5
0	0	0	0	0	0	0
1	1	1	1	1	1	1
2	3	3	4	2	5	3
3	0	1	3	5	2	4
4	0	0	1	0	1	1
5	2	2	5	4	3	2

$N = 5$: $S \not\Rightarrow D$ Counterexample (tight). Roles: $0 = z_1$, $1 = z_2$, $2 = s = r$ (identity on core, pure non-classifier), $3 =$ classifier, $4 =$ MIXED ($4 \cdot 2 = 0$ but $4 \cdot 3 = 2$). Tight: a non-vacuous dichotomy failure needs 3 distinct core roles. Lean: `TightWitnesses.lean` (`s_not_implies_d_tight`).

·	0	1	2	3	4
0	0	0	0	0	0
1	1	1	1	1	1
2	0	2	2	3	4
3	0	0	0	1	0
4	0	0	0	2	2

$N = 5$: $S \not\Rightarrow C$ Counterexample (structural, tight). Roles: $0 = z_1$, $1 = z_2$; core rows are the transpositions $L_2 = (3\ 4)$, $L_3 = (2\ 3)$, $L_4 = (2\ 4)$; every core element is its own retraction pair. No composition of two distinct transpositions is a transposition, so no ICP triple exists. Lean: `TightWitnesses.lean` (`s_not_implies_icp_structural_tight`).

·	0	1	2	3	4
0	0	0	0	0	0
1	1	1	1	1	1
2	0	0	2	4	3
3	0	0	3	2	4
4	0	0	4	3	2

$N = 5$: S+D $\not\Rightarrow$ C Counterexample (tight). Roles: $0 = z_1$, $1 = z_2$, $2 = s = r$ (identity on core), $3 =$ classifier, $4 =$ non-classifier (constant 2 on core). A full `DichotomicRetractMagma` with one classifier and two non-classifiers: ICP is structurally impossible. Lean: `TightWitnesses.lean` (`sd_not_implies_icp_tight`).

·	0	1	2	3	4
0	0	0	0	0	0
1	1	1	1	1	1
2	0	2	2	3	4
3	0	0	0	1	0
4	0	4	2	2	2

$N = 8$: $S \not\Rightarrow D$ Counterexample (superseded; separated roles). Roles: $0 = z_1$, $1 = z_2$, $2 = s$, $3 = r$, $4 =$ classifier, $5, 7 =$ partially classifying (rows on core mix $\{z_1, z_2\}$ and non-absorber outputs, violating Definition 2.4 clause (ii)). Lean: `Countermodel.lean`.

·	0	1	2	3	4	5	6	7
0	0	0	0	0	0	0	0	0
1	1	1	1	1	1	1	1	1
2	3	3	7	3	4	6	5	2
3	0	1	7	3	4	6	5	2
4	0	0	0	0	0	0	1	0
5	6	2	6	2	1	1	1	1
6	0	0	5	2	2	2	2	2
7	2	2	2	1	2	2	6	3

$N=6$: $S \not\Rightarrow C$ Counterexample (structural; superseded by the tight $N=5$ witness, retained for its 4-element core). Roles: $0 = z_1$, $1 = z_2$, $2 = s = r$ (involution on core). Retraction pair holds; no triple satisfies ICP despite 4 core elements. Lean: `E2PM.lean` (`sNoH6_e2pm`).

·	0	1	2	3	4	5
0	0	0	0	0	0	0
1	1	1	1	1	1	1
2	0	3	3	2	5	4
3	2	4	5	5	1	4
4	5	3	0	0	3	2
5	4	2	2	2	2	2

$N=10$: $D \not\Rightarrow C$ Counterexample. Roles: $0 = z_1$, $1 = z_2$, $2 = s$, $3 = r$, $4 = \tau$. No triple satisfies ICP. Lean: `Countermodels10.lean` (`dNotH`), `ICP.lean` (`dNotH_no_icp`).

·	0	1	2	3	4	5	6	7	8	9
0	0	0	0	0	0	0	0	0	0	0
1	1	1	1	1	1	1	1	1	1	1
2	3	3	2	3	4	5	6	7	9	8
3	0	1	2	3	4	5	6	7	9	8
4	0	0	1	1	1	1	1	1	1	1
5	1	0	0	0	0	0	0	0	0	0
6	2	3	9	9	9	9	9	9	9	8
7	3	2	9	9	9	9	9	9	9	8
8	1	0	1	0	1	1	1	1	0	0
9	0	1	0	1	0	1	1	0	1	1

$N = 10$*: $C \nRightarrow D$ Counterexample.* Roles: $0 = z_1$, $1 = z_2$, $2 = s$, $3 = r$, $4 = \tau$. ICP holds ($a = 8, b = 6, c = 7$). Element 5 violates dichotomy. Lean: `Countermodels10.lean` (`hNotD`).

·	0	1	2	3	4	5	6	7	8	9
0	0	0	0	0	0	0	0	0	0	0
1	1	1	1	1	1	1	1	1	1	1
2	3	1	3	4	9	6	8	5	7	2
3	0	1	9	2	3	7	5	8	6	4
4	0	0	1	1	1	1	1	1	0	0
5	0	0	2	0	0	0	0	0	3	1
6	2	2	2	8	3	9	4	7	9	7
7	8	3	2	8	3	9	4	7	3	1
8	9	2	2	3	8	1	3	7	1	7
9	2	2	2	2	4	7	6	7	2	0

$N = 4$*: $D \nRightarrow S$ Counterexample (tight).* Element 2 is a classifier: row on core is $(1, 1) \subseteq \{z_1, z_2\}$. Element 3 is a non-classifier: row on core is $(2, 2) \subseteq$ core. No retraction pair exists among the 4 candidate (s, r) assignments. Tight: D requires $N \geq 4$. Lean: `E2PM.lean` (`d_not_implies_s_tight`).

·	0	1	2	3
0	0	0	0	0
1	1	1	1	1
2	0	1	1	1
3	2	3	2	2

$N = 5$*: $C \nRightarrow S$ Counterexample (tight).* ICP holds but no retraction pair exists. Tight: ICP requires $N \geq 5$. Lean: `E2PM.lean` (`h_not_implies_s_tight`).

·	0	1	2	3	4
0	0	0	0	0	0
1	1	1	1	1	1
2	3	1	0	3	1
3	2	4	3	4	2
4	2	2	1	0	3

$N = 5$*: $C \nRightarrow D$ Counterexample (tight).* ICP holds but the dichotomy fails: no element maps core entirely to $\{z_1, z_2\}$ (no classifier exists). Tight: ICP requires $N \geq 5$. Lean: `E2PM.lean` (`h_not_implies_d_tight`).

·	0	1	2	3	4
0	0	0	0	0	0
1	1	1	1	1	1
2	3	3	4	3	3
3	2	4	4	4	3
4	2	2	2	4	4

$N = 10$*: S+D+C Coexistence Witness (all roles distinct).* S+D+C roles: $0 = z_1$, $1 = z_2$, $2 = s$, $3 = r$, $4 = \tau$, $8 = \eta$ (ICP a), $6 = g$ (ICP b, core-preserving), $7 = \rho$ (ICP c). Lean: `Witness10.lean`.

·	0	1	2	3	4	5	6	7	8	9
0	0	0	0	0	0	0	0	0	0	0
1	1	1	1	1	1	1	1	1	1	1
2	3	3	4	3	7	5	9	6	8	2
3	0	1	9	3	2	5	7	4	8	6
4	0	0	1	1	1	0	0	0	1	1
5	2	2	7	2	8	9	4	3	4	2
6	0	0	6	4	8	7	3	3	4	9
7	2	2	6	4	8	9	4	3	4	9
8	2	2	4	8	4	3	4	4	8	9
9	3	4	7	3	9	2	2	9	2	3

$N = 5$*: Unsorted S+D+C witness.* Roles: $0 = z_1$, $1 = z_2$, $2 = s = r$ (involution $(2\ 3)$ on core, non-classifier), $3, 4 =$ classifiers (ICP a, c). Not sorted: $2 \cdot 3 = 2 \in N$ but $2 \cdot 4 = 4 \in C$, though $3, 4$ share a class. Lean: `Sorting.lean` (`sorting_independent`).

·	0	1	2	3	4
0	0	0	0	0	0
1	1	1	1	1	1
2	0	2	3	2	4
3	0	0	1	0	0
4	0	1	0	1	0

$N = 4$*: Constant-N sorted witness (no retraction pair).* Roles: $0 = z_1$, $1 = z_2$, $2 =$ classifier, $3 =$ non-classifier sending both classes to N. The constant class-table forces $\neg S$ (Proposition 8.4). The constant-C table is realized by `dNoS4` (above). Lean: `Sorting.lean` (`constN4_constN`, `constN4_no_retract`).

·	0	1	2	3
0	0	0	0	0
1	1	1	1	1
2	0	1	1	1
3	0	3	3	3

$N = 6$*: The canonical homoiconic kernel.* Roles: $0, 1 =$ halt states, $2 =$ `quote` (the 4-cycle $(2\ 5\ 3\ 4)$ on core), $3 =$ `eval` (its inverse), $4 =$ `data?` (sort-introspection κ), $5 =$ `judge?`. Sorted, class-swapping, with `judge?` $=$ `data?` $\circ$ `quote` as the ICP triple $(5, 2, 4)$. Lean: `Homoiconic.lean` (`canonical_kernel`).

·	0	1	2	3	4	5
0	0	0	0	0	0	0
1	1	1	1	1	1	1
2	1	0	5	4	2	3
3	0	0	4	5	3	2
4	0	0	1	1	0	0
5	0	0	0	0	1	1

$N = 8$*: The canonical Stack-A artifact.* Roles: $0, 1 =$ halt states, $2 =$ `quote`, $3 =$ `eval`, $4 =$ `shift`, $5 =$ `data?`, $6 =$ `judge?`, $7 =$ `shift?`. Lexicographically minimal member of the 228-table law-determined design space. Lean: `ArtifactN8.lean` (`canonical_artifact_N8`).

·	0	1	2	3	4	5	6	7
0	0	0	0	0	0	0	0	0
1	1	1	1	1	1	1	1	1
2	0	0	5	6	7	2	3	4
3	0	1	5	6	7	2	3	4
4	0	0	5	7	6	2	4	3
5	0	0	1	1	1	0	0	0
6	0	0	0	0	0	1	1	1
7	0	0	0	0	1	0	0	1

$N = 6$*: Class-swapping sorted S+D+C witness.* Roles: $0 = z_1$, $1 = z_2$, $2 = s$, $3 = r$ (pairing $(2\ 4)(3\ 5)$: the section exchanges the non-classifier block $\{2,3\}$ with the classifier block $\{4,5\}$), $4, 5$ = classifiers (ICP a, c). Sorted, in the class-swapping world. Lean: `Sorting.lean` (`swap_world_inhabited`).

·	0	1	2	3	4	5
0	0	0	0	0	0	0
1	1	1	1	1	1	1
2	0	2	4	5	2	3
3	0	3	4	5	2	3
4	0	0	1	0	0	0
5	0	1	0	0	1	0

B PROOF INVENTORY

All 37 Lean files compile with `lake build`, zero `sorry`. Proof styles: *Algebraic* = pure equational reasoning (no `decide`); *decide* = verified by kernel computation ($N \leq 8$); *native_decide* = verified by compiled native code ($N = 10$ and permutation searches), which additionally trusts the Lean compiler (`Lean.ofReduceBool`). Algebraic and `decide` proofs depend on `propext` at most; files marked † use `Classical.choice` and `Quot.sound` through Mathlib's finiteness machinery. The files below the rule constitute the machine ladder developed in the companion write-up; they are proved in this repository and treated there as imported results.

File	**Thms**	**Style**	**Content**
Dichotomic	20	Algebraic	Decomposition, bounds, dichotomy bo
NoCommutativity	4	Algebraic	Asymmetry
OneSidedSeparation	3	decide	One-sided vs mutual-inverse retractio
Functoriality	4	Algebraic	Decomposition invariance
CapabilityInvariance	4	Algebraic	S, D, C each invariant
ICP	20	Alg.+native	ICP ⇔ Compose+Inert
DStruct	1	Algebraic	Axiom reduction: D_{struct} + ICP
Countermodel	5	decide	$S \not\Rightarrow D$ ($N = 8$; superseded)
Countermodels10	9	native_dec.	$D \not\Rightarrow C$, $C \not\Rightarrow D$ ($N = 10$)
E2PM	19	decide	$D \not\Rightarrow S$, $C \not\Rightarrow S$, $C \not\Rightarrow D$ (tight)
TightWitnesses	18	decide	$S \not\Rightarrow D$, $S \not\Rightarrow C$, S+D $\not\Rightarrow$ C (tight, $N = 5$
Witness5	3	decide	S+D+C at $N = 5$, no ICP at $N = 4$
Witness6	3	decide	S+D+C at $N = 6$ ($s \neq r$)
Witness10	6	native_dec.	S+D+C at $N = 10$
WitnessAllN	25	Algebraic	S+D+C at every $N \geq 5$
IndependenceAllN	36	Algebraic	All six non-implications, all 8 cells, ev
CompletenessWall	7	Algebraic†	K-infinity; completeness excludes D
Sorting	28	Alg.+decide	Involution, four class-tables, balance, a
Homoiconic	14	Alg.+decide	Introspection fixes the quotation law;
ArtifactN8	16	decide	Canonical $N = 8$ model: kernel + hygie
StackAForced	4	Algebraic†	Frame derived: swap world forced, N
EvalSideFree	8	Algebraic†	Eval side of the law set free (Thm 8.9)
QuoteOrbit	7	Algebraic†	Finite order, even order, orbit closure (
BooleanCube	25	decide+nat.	Joint irredundance: all 8 cells of $(S, D,$
Rigidity	6	decide+nat.	Role rigidity of principal witnesses
RigidityPartial	1	Algebraic	Partial rigidity maps
StructureN5	8	Algebraic†	$N = 5$ structure theorem: role lock-in
MirrorRow	1	Algebraic†	$N = 5$ automorphisms: $\lvert\mathrm{Aut}\rvert \leq 2$
SelfSimulation	5	Algebraic	Partial application injectivity (suppl.)
KameaRef	1	Algebraic	Reference oracle for the Rust host (sup
Factorization	7	Alg.+decide	Driver metacircularity, minimal form
FactorizationEnv	10	Alg.+decide	+ environments (two-argument eval)
FactorizationClos	15	Alg.+decide	+ closures, fuel, certified divergence
FactorizationCtrl	20	Alg.+decide	+ control: System L machine, μ
FactorizationStore	19	Alg.+decide	+ store: CESK, lockstep bisimulation
FactorizationData	27	Alg.+decide	+ pairs, dispatch, certified homoiconic
FactorizationEqv	26	Alg.+decide	+ `eqv?`: atomic identity, `null?`
Total	**435**		